



Faculty of Engineering and Technology

**Department of Computer Science and Engineering
(CSE)**

Curriculum for B.Sc. Engineering

Session 2020-2021

Examinations

2021 [Year-1, Term-1 & Term-2]

2022 [Year-2, Term-1 & Term-2]

2023 [Year-3, Term-1 & Term-2]

2024 [Year-4, Term-1 & Term-2]

DEPARTMENT AT A GLANCE

The Department of Computer Science and Engineering (CSE) was founded in 2008. The goal of CSE Department is to cultivate highly-motivated and well-trained professionals who will lead the IT area. The Department of Computer Science and Engineering offers various specialized educational programs to create many competent engineers with profound knowledge of academic theories and practical approaches for the development of our country and all human society, in general. The department offers both basic and advanced courses. In the Department of Computer Science and Engineering, students study basic and applied technologies related to CSE, information technology as well as information processing, information systems, and the diverse technologies upon which our IT society is based on. To become engineers with knowledge related to the construction and management of communication networks which serve as transmission media, software driven management, and the control of systems. They support these networks, and knowledge related to hardware design and manufacture. The department has a number of well-constructed laboratories, namely Software Laboratory, Electrical & Electronics laboratory, Communication laboratory. Well-equipped computers are provided for the students, faculty members as well as the researchers. The department has a seminar library. Students are encouraged for academic excellence by awarding various prizes, medals and certificates in per year performances. The department also arranges co-curriculum activities among the students such as programming contests, software exhibitions, cultural events, games competitions, debate etc. in every year.

PREAMBLE:

Pabna University of Science and Technology (PUST) have introduced B.Sc. Engineering degree from the session 2008-09 under the Faculty of Engineering and Technology. This program of study shall extend over a period of 04 (four) academic years which shall consist of 08 (eight) terms. The credit point system will be the deciding factor to assess this program. All the departments under this faculty will have full autonomy to develop guidelines and conduct all types of academic activities within it strictly observing this ordinance.

1. Departments

The University shall have the following undergraduate degree awarding departments in the five Faculties:

(a) Faculty of Engineering and Technology

- (i) Department of Computer Science and Engineering (abbreviated as CSE)
- (ii) Department of Electrical and Electronic Engineering (abbreviated as EEE)
- (iii) Department of Electrical, Electronic and Communication Engineering (abbreviated as EECE)
- (iv) Department of Information and Communication Engineering (abbreviated as ICE)
- (v) Department of Civil Engineering (abbreviated as CE)
- (vi) Department of Architecture
- (vii) Department of Urban and Regional Planning (abbreviated as URP)

(b) Faculty of Science

- (i) Department of Mathematics
- (ii) Department of Physics
- (iii) Department of Pharmacy
- (iv) Department of Chemistry
- (v) Department of Statistics

(c) Faculty of Business Studies

- (i) Department of Business Administration (abbreviated as B.Admin)
- (ii) Department of Tourism and Hospitality Management (abbreviated as THM)

(d) Faculty of Humanities and Social Science

- (i) Department of Economics
- (ii) Department of Bengali
- (iii) Department of Social Work (abbreviated as SW)
- (iv) Department of English
- (v) Department of Public Administration
- (vi) Department of History and Bangladesh Studies (abbreviated as HBS)

(e) Faculty of Life and Earth Science

- (i) Department of Geography and Environment

And any other Institute/Faculty/Department which may be approved by the University authority from time to time.

2. Degree Offered

The University shall offer courses leading to the award of the following undergraduate degrees in the five Faculties:

(a) Faculty of Engineering and Technology

- (i)** Bachelor of Science in Engineering, abbreviated as B. Sc. Engineering.
- (ii)** Bachelor of Science in Architecture, abbreviated as B. Arch.
- (iii)** Bachelor of Science in Urban and Regional Planning, abbreviated as BURP.

(b) Faculty of Science

- (i)** Bachelor of Science (Honours), abbreviated as B. Sc. (Honours)
- (ii)** Bachelor of Pharmacy (Professional), abbreviated as B. Pharm. (Professional)

(c) Faculty of Business Studies

- (i)** Bachelor of Business Administration, abbreviated as BBA
- (ii)** Bachelor of Business Administration in Tourism and Hospitality Management, abbreviated as BBA (THM)

(d) Faculty of Humanities and Social Science

- (i)** Bachelor of Social Science (Honours), abbreviated as BSS (Honours)
- (ii)** Bachelor of Arts (Honours), abbreviated as BA (Honours)

(e) Faculty of Life and Earth Science

Bachelor of Science (Honours), abbreviated as B. Sc. (Honours)

Any other degree that may be awarded by the Department on the approval of the Regent Board on the recommendation of the Faculty and the Academic Council.

3. Admission and Re-admission

Students shall be enrolled into the first year first semester of the B.Sc. Engineering in the department of Computer Science and Engineering as per the University Rules and Act.

a) Eligibility

Students passing the Secondary School Certificate (SSC) examination and Higher Secondary Certificate (HSC) examination or a recognized equivalent examination in Bangladesh or abroad may be admitted to the B. Sc. Engineering programme of this University on such terms and conditions as may be determined by the University admission committee constituted by the competent authority.

b) Admission Cancellation

- (i) If a newly admitted student remains totally absent without prior permission of the Registrar through the Chairman of the Department from all classes for first 02 (two) consecutive weeks after the start of first year first semester classes, his/her admission shall be cancelled on the recommendation of the Chairman of the concerned Department.
- (ii) A student shall never take admission simultaneously in more than one subject of this University or in any other higher institution with an exception of Certificate/Diploma course. If the stated clause is violated, studentship, examination, and examinations' results of the reported student shall be cancelled.

c) Re-admission

- (i) A student who fails to earn YGPA (including 1st and 2nd semester) less than 2.00, unless otherwise the section 4.b (i) is applicable, may be allowed to get re-admission with the 1st/2nd semester of the immediate next batch. Readmitted students, however, shall always be assigned the original registration number.
- (ii) If a student fails to appear (not allowed to fill the examination entry form) at any semester final examination (other than first year first semester) due to shortage of required percentage of attendance, or failure of paying the dues of the University or for any other reason s/he shall have to get herself/himself re-admitted to the same semester of the subsequently available batch.
- (iii) If a student fails to get the requisite grade points for promotion (according to clause 16) from one year to the next, may seek re-admission with the 1st/2nd semester of the subsequent available batch.
- (iv) On re-admission, grades earned earlier by a student in any semester shall be cancelled automatically and the student shall have to retake all the course-works (such as class test/quiz test, internship, Viva-voce, final examination etc.) of that semester. Percentage of class attendance of such students shall be counted from the date of her/his re-admission. Class test(s) if completed before her/his re-admission, the concerned course teacher shall arrange make-up examination(s).
- (v) A student shall not get chance for re-admission more than three times during the entire program.
- (vi) For re-admission, a student shall have to apply within 02 (two) consecutive weeks after announcement of the result of the concerned semester.

4. Registration

A student shall attain her/his studentship for the University to an academic programme as per the University rules. S/he shall be required to register with the

University through the University registration process on payment of required fees as determined by the University authority from time to time.

5. Medium of Instruction

The medium of instruction and answer in the examination script for the programmes mentioned in the section 2 shall be in English.

6. Programme Duration

Each academic year is divided into two semesters to be called as first semester and second semester. Distribution of weeks in each semester shall be as follows:

(i)	Class Teaching	14 weeks
(ii)	Sessional/Laboratory Examination	01 week
(iii)	Preparation Time for Semester Final Examination	02 weeks
(iv)	Semester Final Examination	03 weeks
Total=		20 weeks

Normally 01 (one) week break is provided after semester final examination.

7. Syllabus/Curriculum

There shall be a syllabus/curriculum of the programme for each batch prepared by the committee of courses as constituted by the University. The syllabus/curriculum should clearly indicate the courses, and detail course contents, credit points and total number of credits in each semester and year. The syllabus/curriculum shall be updated by the concerned committee of courses as and when needed by the academic committee of the concerned Department.

8. Structure of Courses

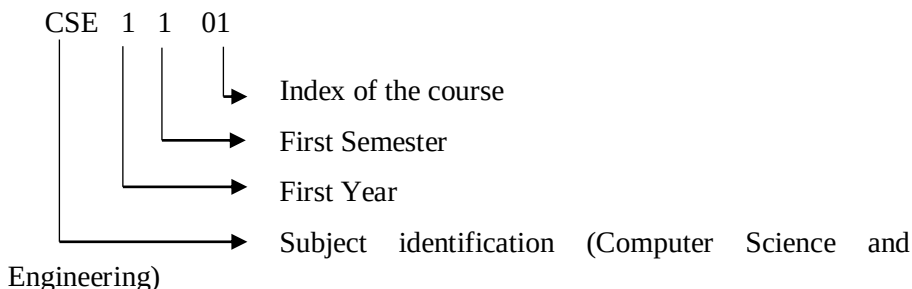
Syllabus/curriculum of each Department in the five Faculties of this University shall consist theoretical, sessional/laboratory, project and or thesis works, internship and Viva-voce etc.

Course Identification

Each course is designed by three to four letter words identifying the subject followed by a four digit number as follows:

- (i) The first digit corresponds to the 'year' indicated in which the course is taken by the student;
- (ii) The second digit corresponds to the 'semester' in which the course is taken by the student;
- (iii) The third and fourth digits shall be used to identify a 'course' within the Department.

For Example:



9. Credit Requirements and Contact Hours

(a) B.Sc. (Engineering) Degree

The total credits for the 04 (four) years B. Sc. (Engineering) degree and the distribution will be decided by the individual Departments. However, the following constraints are operative.

- (i) Total Credits: 150-165
- (ii) Maximum number of credits for sessional courses/projects/field study: 48
- (iii) Maximum credits for Viva-voce: 16
- (iv) There shall be at least 18 credits including Viva-voce in each semester.

(b) Contact Hours

There shall be at least 01 (one) lecture-hour for each credit in a week for each theoretical course and 02 (two) practical-hour for each credit in a week for each sessional/laboratory course.

10. Entry Requirements for Examination

- (i) A student shall be allowed to appear at the semester final examination as a regular student if her/his class attendance is at least 70% of the total courses held in a semester.
- (ii) Student having 60-69% attendance are considered to be non-collegiate and will be eligible to sit for the final examination on payment of fine Tk. 1000/- (One thousand) to the respective department.
- (iii) A student shall not be allowed to appear at the examination if her/his class attendance is below 60% of the total courses held in a Semester.
- (iv) A student is required to fill the examination entry form and pay the University dues within the time specified by the concerned authority for taking each semester final examination. The Chairman of the Department

shall send these examination entry forms to the Controller of Examinations.

11. Duration of Final Examination

- (i) The semester final examination for each theory course shall be of 03 (three) hours duration. There shall be two examiners for each course in the semester final examination.
- (ii) Duration of each sessional/laboratory examination will be between 4-6 hours irrespective of credits.

12. Evaluation System

(a) Each course offered by the Department should be composed of 100 marks. The marking and student evaluation system will be as follows:

(i) Distribution of marks for each theoretical course

Thirty percent (30%) of marks of a theoretical course shall be allotted for continuous assessment, i.e. quizzes, class test, home assignments, class evaluation and class performance. The rest of the marks (seventy) will be allotted to the semester final Examination that is conducted centrally by the University. Distribution of marks for a given course is as follows.

Class Attendance	10%
Class Tests/Assignments/Quizzes	20%
Final Examination (3hours)	70%
Total	100

The number of class tests/ assignments/quizzes of a course shall be three. Evaluation of performance in the class tests/ assignments/quizzes will be on the basis of the best two class tests/ assignments/quizzes of the three class tests/assignments/quizzes. The class test, assignment, quiz etc. will be conducted and evaluated by the course teacher and the semester final scripts will be evaluated by two (three, if needed) examiners.

(ii) Distribution of marks for sessional/laboratory course

Forty percent (40%) of marks of a sessional/laboratory course shall be allotted for continuous assessment, i.e. lab test, quizzes, home assignments, lab evaluation and lab performance. The rest of the marks (sixty) will be allotted to the semester final sessional/laboratory examination. Distribution of marks for the sessional/laboratory courses is as follows:

Sessional/Laboratory Attendance	10%
Lab Test/Lab Performance and Lab Report	30%
Final Sessional/Laboratory Examination	60%
Total=	100%

Each department shall have one external examiner (not below the rank of Assistant Professor) for any one of the sessional/laboratory examinations in each semester.

(iii) Project/Thesis: Project/Thesis marks to be evaluated as follows:

Presentation and Viva-voce (Conducted by the Examination Committee)	20%
Assessment of Supervisor	30%
Project/Thesis Examined by Two Examiners* (Average of the marks given by the two examiners shall be taken as final)	50%
Total=	100%

*Two examiners to evaluate the project/thesis shall be selected by the concerned examination committee of the Department.

(iv) Class Attendance: The distribution of marks for class attendance (theoretical and sessional/laboratory) shall be as follows:

Attendance	Marks
90% and above	10
85% to less than 90%	9
80% to less than 85%	8
75% to less than 80%	7
70% to less than 75%	6
60% to less than 70%	5
Below 60%	0

b) Continuous Assessment Report:

- (i) At the end of the course, the course teacher shall calculate total marks of the continuous assessment (including class attendance) and prepare a marks sheet. S/he shall notify it to the students and submit the same to the Chairman of the examination committee as well as to the Controller of Examinations before start of the semester final examination.

- (ii) The course teacher shall also submit the class attendance report with the register/ documents to the Chairman of the Department for percentage calculation before sending the examination entry forms to the Controller of Examinations.

c) First and Second Examiners: There shall be two question setters and examiners (first and second) for each course of the semester final examination. The course teacher shall act as first examiner. The examination committee shall select the second examiner from the “Panel of Examiners” as approved by the Academic Council.

d) Final Marks: The average of the marks given by the two examiners shall be taken as final. If the marks of two examiners are differed by 20% or more, the relevant answer script (s) will go for third examination. In that case, the average of nearest two marks will be taken as final marks. If two marks are equidistant from the other, then average of three marks will be taken as the final.

e) Submission of Mark Sheets: All examiners including teachers involved in continuous assessment, Viva-voce, etc., shall prepare 04 (four) copies of mark sheets and submit 03 (three) copies to the Chairman of the respective examination committee and 01 (one) copy to the Controller of Examinations. Total marks in each course (final examination/tutorial assignment/attendance) should be rounded-up and awarded only one grade.

f) Letter Grade and Grade Point: Total marks obtained in each course, oral (viva-voce) examination, internship etc. shall be converted into letter grade (LG) and grade point (GP) as follows:

Numerical Grade	Letter Grade		Grade point	Interpretation
80% and above	A+	(A plus)	4.00	Outstanding
75% to less than 80%	A	(A regular)	3.75	Excellent
70% to less than 75%	A-	(A minus)	3.50	Very Good
65% to less than 70%	B+	(B plus)	3.25	Good
60% to less than 65%	B	(B regular)	3.00	Satisfactory
55% to less than 60%	B-	(B minus)	2.75	Below Satisfactory
50% to less than 55%	C+	(C plus)	2.50	Average
45% to less than 50%	C	(C regular)	2.25	Pass
40% to less than 45%	D		2.00	Poor
less than 40%	F	...	0.00	Fail

g) Re-examination: Re-examination of any script shall not be allowed.

13. Result Publication

a) The following information shall be shown in the tabulation sheet.

- (i) **Semester final result preparation:** Continuous assessment marks, semester final marks, total marks, LG and GP in each course and earned credit points secured, total credits, Grade Point Average (GPA), Yearly Grade Point Average (YGPA), Cumulative Grade Point Average (CGPA) till the semester will be calculated and placed in the tabulation book.
- (ii) **Retake/Improvement result preparation:** The retake/improvement marks of the students, if any, shall be written in the original tabulation sheet by the concerned examination committee;
- (iii) **Manifestation in semester final result sheet:** The semester final result sheet, which shall be published for the students, shall not show the numerical marks but shall show the LG, GP in each course, GPA and CGPA. If a student get 'F' grade in any course, it will remain permanently on the grade sheet and his/her final transcript.

b) The Controller of Examinations shall publish the results of each semester, year and the whole programme, subject to the approval of the Vice-Chancellor/Regent Board and thereafter shall send copy of the tabulation sheets duly signed by him with date to the Chairman of the concerned Department. The Controller of Examination shall also provide the transcript/grade sheet showing course-wise LG and the corresponding GP (the numerical marks shall not be shown), the CGPA, LG and the interpretation of the CGPA of the candidates. Final grade must be spelled out clearly in the certificate/ transcript.

[Example: C+= 'C+' (C plus); A-= 'A-' (A minus); B= 'B' (B regular)].

- (i) The result of the semester final examinations (except final year second semester) shall be published **within 03 (three) weeks** from the date of last examination.
- (ii) Result sheets of each examination as prepared by the Controller of Examinations shall be compared and signed by at least two tabulators.

14. Promotion

- (i) All promotions from first year first semester to final year second semester shall be year based.
- (ii) For promotion to the next year, a student has to score at least YGPA 2.00.
- (iii) A student obtaining 'F' grade in any course shall be allowed to improve only with the available batch/batches within her/his total course duration.

15. Degree Requirements

For the B. Sc. Engineering degree, each student shall require to:

- (i) Earn the required number of total credit points successfully;
- (ii) Earn a minimum CGPA of 2.00;
- (iii) Complete the programme within 07 (seven) academic years of his/her first admission year into the programme; and

16. Improvement of Grades

- (i) If a student obtains a grade lower than B- (B minus) in the courses allotted in any year, s/he will be allowed to repeat to sit for maximum 50% courses only once with the following batch for the purpose of grade improvement. A student failing to improve his or her grade in a course can retain the earlier grade.
- (ii) **Improvement in final year first and second semester courses:** Student would be allowed to sit for improvement examination in the final year first and second semester courses only with the following batch.

17. Drop Out

- (i) If a three times re-admitted student in any semester failed to earn minimum required GPA for promotion s/he shall be dropped out from the programme.
- (ii) For the degrees of B. Sc. Engineering, if it seems that it is not possible for a student to complete the programme within 07 (seven) academic years (14 Semesters), s/he shall be dropped out from the programme.
- (iii) For the degrees of B. Sc. Engineering, if a student fails to earn required total credit points within 07 (seven) academic years since first admission, s/he will be dropped-out from the programme.

18. Credit Transfer

No credit transfer from any other Programmes/Universities/ Institutions to the Pabna University of Science and Technology shall be allowed for the programme.

19. Academic Administration

- a) **Academic Calendar:** The academic calendar prepared by the Dean of the Faculty showing dates of beginning and closing classes, commencement of examinations and probable dates for publication of the result shall be published by the respective Departmental academic committee before commencement of each semester. The copy shall be sent to the Controller of Examinations and the respective University authority.

- b) **Academic and co-academic activities:** Within the framework of these rules and regulations and the rules of the university, the Departmental academic committee may adopt policies for strengthening the academic and co-academic activities of the Department.

20. Computation of Grade Point Average

(a) Grade Point Average:

A Grade Point Average (GPA) shall be calculated for each semester as follows:

$$GPA = \frac{\sum_{i=1}^n C_i * G_i}{\sum_{i=1}^n C_i} \dots \dots \dots (i)$$

Where, n is the number of courses offered during the semester, C_i is the number of credits allotted to a particular course, and G_i is the grade point earned for that course.

(b) Yearly Grade Point Average:

A Yearly Grade Point Average (YGPA) shall be calculated for each academic year as follows:

$$YGPA = \frac{\sum_{j=1}^2 C_j * G_j}{\sum_{j=1}^2 C_j} \dots \dots \dots (ii)$$

Where, 2 is the number of semester, C_j is the number of credits allotted to a semester and G_j is the GPA earned for that semester.

(c) Cumulative Grade Point Average:

The Cumulative Grade Point Average (CGPA) gives the cumulative performance of the students from the first year up to the end of the year to which it refers, and will be calculated as follows:

$$CGPA = \frac{\sum_{k=1}^m C_k * G_k}{\sum_{k=1}^m C_k} \dots \dots \dots (iii)$$

Where, m is the total number of years being considered, C_k is the total number of credits registered during a year, and G_k is the YGPA of that particular year.

20.1 Numerical Examples of computing GPA and CGPA

20.1.1 Example for Computing GPA

Say, a student completed nine courses in 1st year 1st semester and obtained the following grade:

Course	Credit	Grade (C_i)	Grade Points (G_i)	$C_i * G_i$
CSE 1101	3.00	A+	4.00	12.00
CSE 1102	1.50	F	0.00	0.00
CSE 1103	3.00	A	3.75	11.25
CSE 1104	1.50	D	2.00	3.00
MATH 1101	3.00	A+	4.00	12.00
PHY 1101	3.00	A-	3.50	9.75
HUM 1101	3.00	F	0.00	0.00
HUM 1102	0.75	D	2.00	1.50
CSE 1150	0.75	D	2.00	1.50
Total	19.50			51.00

Then, the GPA of the semester is calculated as follows:

$$\text{GPA} = 51.00/19.50 = 2.615 \approx 2.62$$

20.1.2 Example for Computing YGPA

Say, a student completed two semesters in a year and obtained the following GPAs

Year	Semester	Credit C_j	GPA Earned, G_j	$G_j * C_j$
First	First	19.50	2.615	51.00
First	Second	19.50	3.56	69.42
Total		39.00		120.42

Then, the YGPA is calculated as follows:

$$\text{YGPA} = 120.42/39.00 = 3.087 \approx 3.09$$

20.1.3 Example for Computing CGPA

Say, a student completed four semesters and obtained the following GPAs

Year	Semester	Credit, C_k	GPA Earned, G_k	$G_k * C_k$
First	First	19.50	2.615	51.00
First	Second	19.50	3.56	69.42
Second	First	20.25	3.40	68.85
Second	Second	19.25	4.00	77.00
Total		78.5		266.27

Then, the CGPA is calculated as follows:

$$\text{CGPA} = 266.27 / 78.50 = 3.391 \approx 3.39$$

21. Dean's List

As a recognition of excellent academic performance, the name(s) of student(s) obtaining an average CGPA of 3.80 or above (without any improvement) shall be published after completion of the undergraduate programmes in the Dean's List in each Faculty. Students penalized/punished for any offense will not be considered for the Dean's List. Such a student will not be considered for any other academic rewards also.

22. Amendment of the Ordinance

This Ordinance may be amended by the Academic Council on the recommendation of the five Faculties of the University.

B.Sc. Engineering in Computer Science and Engineering

The Computer Science and Engineering programme combines an objective and realistic education in computer science with added emphasis on the physical and architectural underpinnings of state-of-the-art computing systems. With a strong background in both computer science and computer engineering, our graduates are able to fit in any computing environment across the hardware-software spectrum.

Vision

To create the most conducive environment for academic excellence and research-based education in the field of Computer Science and Engineering and hence produce competent computer professionals with sound knowledge of academic research and innovation to compete at the global technological community by excellent industry-academia collaboration, research and education.

Mission

M1. Provide a learning environment to create innovative and problem solving skills with professionalism.

M2. Create facilities and expertise in advanced computer technology to encourage education and research.

M3. Enhance Industry Academia Interaction program to get in touch with industry trends and corporate culture.

M4. Impart moral and ethical values, social commitment and interpersonal skills to the graduates.

The Program Educational Objectives:

PEO1: Work productively or be active in higher education and research as a successful computer professional on a variety of career paths, including supportive and leadership roles in multidisciplinary teams.

PEO2: Communicate effectively, recognize and incorporate social needs and constraints into their professional endeavors, and practice their profession with high respect for ethical responsibilities.

PEO3: Engage in life-long learning and to remain current in their profession to foster personal, industrial and organizational growth.

Mission	Program Educational Objectives		
	PEO1	PEO2	PEO3
M1	√		√
M2	√	√	
M3	√	√	√
M4		√	√

Program Learning Outcomes (PLO):

PLO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals to solve complex problems of computer and information technology
PLO2	Problem analysis	Identify, formulate, research and analyze complex computer engineering problems
PLO3	Investigation	Conduct investigations of complex computer science and technology problems, considering experimental design, data analysis and interpretation.
PLO4	Modern tool usage	Create, select and apply appropriate techniques, resources and modern computer engineering and ICT tools.
PLO5	Design and/or development of solutions	Design and develop solutions for computer engineering problems with components or processes that meet the specified needs with technological concerns
PLO6	The engineer and society	Apply reasoning informed by contextual knowledge to develop the computer technologies for society benefits
PLO7	Environment and sustainability	Understand the impact of professional computer engineering solutions in societal and environmental contexts and demonstrate the knowledge for sustainable development.
PLO8	Ethics	Apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice
PLO9	Individual work and teamwork	Individual work and teamwork: Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings
PLO10	Communication	Communicate effectively about complex computer science and engineering activities with the engineering community and with society at large in both oral and written

PLO11	Project management and finance	Demonstrate knowledge and understanding of computer engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments
PLO12	Life-long learning	Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change

Mapping between Program Educational Objectives (PEO) and Program Learning Outcomes (PLO)

	Program Learning Outcomes											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
PEO1	√	√	√	√	√			√	√	√	√	
PEO2						√	√	√				
PEO3							√	√	√	√	√	√

1. DISTRIBUTION OF UNDERGRADUATE COURSES

Course Type	Credits	Percentage
Mathematics and Basic Sciences	21.00	12.72%
Humanities	11.75	7.12%
Electrical, Electronics, Communication and Mechanical Engineering	14.50	8.79%
Core Courses	117.75	71.37%
Total	165	100%

2. TERM-WISE CREDIT HOURS DISTRIBUTION FOR THE DEGREE OF B.SC. ENGINEERING:

Year	Semester	Credit(s)	
Year-1	Term-1	19.50	39.00
	Term-2	19.50	
Year-2	Term-1	20.25	39.50
	Term-2	19.25	
Year-3	Term-1	21.00	43.00
	Term-2	22.00	
Year-4	Term-1	21.00	43.50
	Term-2	22.50	
Total			165

3. COURSES OFFERED TO THE UNDERGRADUATE STUDENTS DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Year-1, Term-1 B.Sc. Engineering Courses					
S/ N	Course No.	Course Title	Theory H/W	Lab H/W	Credit
1	CSE 1101	Computer Fundamentals	3	0	3
2	CSE 1102	Computer Fundamentals Sessional	0	3	1.5
3	CSE 1103	Structured Programming Language	3	0	3
4	CSE 1104	Structured Programming Language Sessional I	0	3	1.5
5	MATH 1101	Differential Calculus and Co- ordinate Geometry	3	0	3
6	PHY 1101	Physics	3	0	3
7	HUM 1101	Communicative English	3	0	3
8	HUM 1102	Communicative English Sessional	0	3/2	0.75
9	CSE 1150	Viva Voce			0.75
Total			15	9	19.50
Year-1 Term-1 Load = 15L+3.75P+0.75V=19.50 Credits					

Year-1, Term-2 B.Sc. Engineering Courses						
S/ N	Course No.	Course Title	Pre- Requisite	Theory H/W	Lab H/W	Credit
1	ME 1200	Engineering Drawing		0	2	1
2	CSE 1200	Structured Programming Language Sessional II		0	3/2	0.75
3	CSE 1201	Object Oriented Programming	CSE 1103	3	0	3
4	CSE 1202	Object Oriented Programming Sessional		0	3	1.5
5	CSE 1203	Discrete Mathematics		3	0	3
6	EEE 1201	Basic Electrical Engineering		3	0	3
7	EEE 1202	Basic Electrical Engineering Sessional		0	3	1.5
8	MATH 1201	Integral Calculus, Differential Equation and Series Solutions		3	0	3
9	HUM 1201	Economics		2	0	2
10	CSE 1250	Viva Voce				0.75
Total				14	8	19.50
Year-1 Term-2 Load = 14L+4.75P+0.75V=19.50 Credits						

Year-2, Term-1 B.Sc. Engineering Courses						
S/ N	Course No.	Course Title	Pre- Requisite	Theory H/W	Lab H/W	Credit
1	CSE 2101	Data Structures	CSE 1103	3	0	3
2	CSE 2102	Data Structures Sessional		0	3	1.5
3	CSE 2103	Design Pattern and Java Programming	CSE 1201	3	0	3
4	CSE 2104	Design Pattern and Java Programming Sessional		0	3	1.5
5	EEE 2101	Electronic Devices and Circuits		3	0	3
6	EEE 2102	Electronic Devices and Circuits Sessional		0	3	1.5
7	MATH 2101	Vector, Matrices and Linear Algebra		3	0	3
8	STAT 2101	Elementary Statistics and Probability		3	0	3
9	CSE 2150	Viva Voce				0.75
Total				15	9	20.25
Year-2 Term-1 Load = 15L+4.5P+0.75V=20.25 Credits						

Year-2, Term-2 B.Sc. Engineering Courses						
S/ N	Course No.	Course Title	Pre- Requisite	Theory H/W	Lab H/W	Credit
1	CSE 2200	Analytical Programming Sessional	CSE 1103 CSE 1201	0	3	1.5
2	CSE 2201	Algorithms	CSE 2101	3	0	3
3	CSE 2202	Algorithms Sessional		0	3	1.5
4	CSE 2203	Theory of Computation		2	0	2
5	CSE 2205	Digital Systems		3	0	3
6	CSE 2206	Digital Systems Sessional		0	3	1.5
7	MATH 2201	Complex analysis, Laplace and Fourier Transforms		3	0	3
8	STAT 2201	Theory of Statistics		3	0	3
9	CSE 2250	Viva Voce				0.75
Total				14	9	19.25
Year-2 Term-2 Load = 14L+4.5P+0.75V=19.25 Credits						

Year-3, Term-1 B.Sc. Engineering Courses						
S/ N	Course No.	Course Title	Pre- Requisite	Theory H/W	Lab H/W	Credit
1	CSE 3101	Computer Architecture and Organization	CSE 2205	3	0	3
2	CSE 3103	Compiler Design	CSE 2203	3	0	3
3	CSE 3104	Compiler Design Sessional		0	3	1.5
4	CSE 3105	Numerical Methods		3	0	3
5	CSE 3106	Numerical Methods Sessional		0	3/2	0.75
6	CSE 3107	Database Management Systems		3	0	3
7	CSE 3108	Database Management Systems Sessional		0	3	1.5
8	ECE 3101	Data Communication		3	0	3
9	ECE 3102	Data Communication Sessional		0	3	1.5
10	CSE 3150	Viva Voce				0.75
Total				15	10.5	21
Year-3 Term-1 Load = 15L+5.25P+0.75V=21 Credits						

Year-3, Term-2 B.Sc. Engineering Courses						
S/ N	Course No.	Course Title	Pre- Requisite	Theory H/W	Lab H/W	Credit
1	CSE 3200	Project		0	2	1
2	CSE 3201	System Analysis and Design		3	0	3
3	CSE 3203	Operating Systems		3	0	3
4	CSE 3204	Operating Systems Sessional		0	3	1.5
5	CSE 3205	Web Engineering		3	0	3
6	CSE 3206	Web Engineering Sessional		0	3	1.5
7	CSE 3207	Digital Signal Processing	MATH 2201	3	0	3
8	CSE 3208	Digital Signal Processing Sessional		0	3/2	0.75
9	CSE 3209	Microprocessors and Assembly Language	CSE 3101	3	0	3
10	CSE 3210	Microprocessors and Assembly Language Sessional		0	3	1.5
11	CSE 3250	Viva Voce				0.75
Total				15	12.5	22
Year-3 Term-2 Load = 15L+6.25P+0.75V=22 Credits						

Year-4, Term-1 B.Sc. Engineering Courses						
S/ N	Course No.	Course Title	Pre- Requisite	Theory H/W	Lab H/W	Credit
1	CSE 4100	Project/Thesis		0	3	1.5
2	CSE 4101	Software Engineering		3	0	3
3	CSE 4102	Software Engineering Sessional		0	3/2	0.75
4	CSE 4103	Artificial Intelligence		3	0	3
5	CSE 4104	Artificial Intelligence Sessional		0	3	1.5
6	CSE 4105	Digital Image Processing		3	0	3
7	CSE 4106	Digital Image Processing Sessional		0	3	1.5
8		Optional - I		3	0	3
9	HUM 4101	Sociology and Bangladesh Studies		3	0	3
10	CSE 4150	Viva Voce				0.75
Total				15	10.5	21
Year-4 Term-1 Load = 15L+5.25P+0.75V=21 Credits						

NB: The Courses, CSE 4100 and CSE 4200, will be evaluated combined after completing CSE 4200

List of Optional Courses

Optional-I

Optional-I will be selected from the following courses:

S/ N	Course No.	Course Title	Theory H/W	Lab H/W	Credit
1	CSE 4107	Computer Simulation and Modeling	3	0	3
2	CSE 4109	Multimedia Technology	3	0	3
3	CSE 4111	Basic Graph Theory	3	0	3
4	CSE 4113	Parallel and Distributed Processing	3	0	3

Year-4, Term-2 B.Sc. Engineering Courses						
S/ N	Course No.	Course Title	Pre- Requisite	Theory H/W	Lab H/W	Credit
1	CSE 4200	Project/Thesis	CSE 4100	0	6	3
2	CSE 4201	Computer Networks		3	0	3
3	CSE 4202	Computer Networks Sessional		0	3	1.5
4	CSE 4203	Computer Graphics		3	0	3
5	CSE 4204	Computer Graphics Sessional		0	3	1.5
6	CSE 4205	Interfacing and Microcontrollers	CSE 3209	3	0	3
7	CSE 4206	Interfacing and Microcontrollers Sessional		0	3/2	0.75
8		Optional – II		3	0	3
9	HUM 4201	Industrial Management and Accounting		3	0	3
10	CSE 4250	Viva Voce				0.75
Total				15	13.5	22.5
Year-4 Term-2 Load = 15L+6.75P+0.75V=22.5 Credits						

NB: The Courses, CSE 4100 and CSE 4200, will be evaluated combined after completing CSE 4200.

List of Optional Courses

Optional-II

Optional-II will be selected from the following courses:

S/ N	Course No.	Course Title	Theory H/W	Lab H/W	Credit
1	CSE 4207	Cryptography and Network Security	3	0	3
2	CSE 4209	VLSI Design	3	0	3
3	CSE 4211	Wireless Communication	3	0	3
4	CSE 4213	Computational Geometry	3	0	3

YEAR-1 TERM-1 B. Sc. ENGINEERING COURSES

CSE 1101: Computer Fundamentals	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 1102: Computer Fundamentals Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- Give students an in-depth understanding of why computers are essential components in business, education and society.
- Introduce the fundamentals of computing devices and reinforce computer vocabulary, particularly with respect to personal use of computer hardware and software, the Internet, networking and mobile computing.
- Provide hands-on use of Microsoft Office 2013 applications Word, Excel, Access and PowerPoint. Completion of the assignments will result in MS Office applications knowledge and skills.
- Provide foundational or “computer literacy” curriculum that prepares students for life-long learning of computer concepts and skills. Completion of course fulfills pre-requisite to enroll in other computer science courses required for a certificate, A.A. degree, and transfer to the university or to qualify for high demand employment.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Bridge the fundamental concepts of computers with the present level of knowledge of the students.
CO 2	Familiarize operating systems, programming languages, peripheral devices, number systems, logic circuits, multimedia and internet
CO 3	Identify categories of programs, system software and applications. Organize and work with files and folders.
CO 4	Recognize various types of networks, network standards
CO 5	Understand computer ethics
CO 6	Understand cyber security and cyber law

CO No	Program Learning Outcomes (PLO)											
	PLO1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√	√		√								
CO2	√	√	√									
CO3	√			√								
CO4	√			√								
CO5							√	√				√
CO6								√				√

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction to Computers; History and development of Computers; Generation of Computers; Types of Computers.	CO 1	3	3
Basic Units of Computer Hardware; Keyboard; Mouse; Internal structure of CPU; Functions of RAM; ROM and Cache memory; Basic functional mechanism of HDD and CD-ROM; Different types of Monitors; Impact and Non-impact Printers; Scanner; Plotter; Typical Computer specifications. Machine language; Assembly language; High level language; Assembler; Translator; Interpreter and Compiler.	CO 2	6	6
Classifications of software ; System software; Operating system concepts and importance; Components and basic functions of DOS; Windows operating system; Application software and Utility programs; Computer Virus; Application software: text processing (MS-WORD; etc); Spread sheet (MS-EXCEL etc).	CO 3	12	18
Computer networks and its goals; Basic concepts on LAN; MAN; WAN and Internet systems; Internet services; Functions of Modem in Internet.	CO 4	6	3
Computers in the workplace; Computer crime; Rules of communications; Privacy; Intellectual property.	CO 5	9	6
Jurisdiction and Cyber Crime; Criminal Justice in Bangladesh and Implications on Cyber Crime; Hacking, Malicious Spreading in Viruses; Password fraud, Cheating; Cyber Pornography; Protection of Copyrights and Intellectual Property right, Unsolicited e-Mail, Defamation, Harassment and e-Mail Abuse, Present Legal Protection. Definition Nature of cyber law, Scope, Utility of Cyber Law; Origin and Development of Cyber Law and Internet;	CO 6	6	6

Recommended Books:

1. Deborah G. Johnson ; Computer Ethics
2. Sara Baase : Gift of Fire: Social, Legal and Ethical Issues for Computing Technology
3. V.Rajaraman : Fundamentals of Computer

4. Peter Norton : Computing Fundamentals
5. V. D. Dudej: Information Technology & Cyber Laws, *Commonwealth Publishers*
6. Vivck Sood: Cyber Law Simplified, *Tata McGraw Hill Publications*

CSE 1103: Structured Programming Language	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 1104: Structured Programming Language Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- Introduce the students to computer programming using structured language.
- Enhance their analyzing and problem-solving skills and use the same for writing programs using C language.

This course is prerequisite of maximum core course of CSE

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Explain the fundamentals of programming and basic structure of C programming language.
CO 2	Compare and apply the appropriate decision making and control statements, arrays and functions to solve computational problems.
CO 3	Apply appropriate data structures like pointers, structures, unions, user defined data types, and dynamic memory to solve computational problems.
CO 4	Compare and apply language constructs and data structures; perform and demonstrate skills and write report to design, build, and test moderately complex computational problems.

CO No	Program Learning Outcomes (PLO)											
	PLO1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√	√										
CO2	√	√	√	√	√							
CO3	√	√	√	√	√							
CO4		√	√	√	√	√			√			√

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction to computer programming: Algorithm; Writing; debugging and running programs using C	CO 1 CO 2	9	9

compiler. C Basics: Different Data types and their range; Operator and operands and its precedence; Input/Output; Conditional operators; Loops nested structure; Error handling; Built-in functions.			
Functions and Arrays: Writing & calling of User – defined functions; Recursive functions; Scope of variables; Introduction to one-dimensional arrays; Multi-dimensional arrays and array manipulation.	CO 2 CO 3	12	12
Pointers and Strings: Introduction to pointers; Pointers and array; Pointers and functions; String I/O; String-based built-in functions; String operations; Pointer and string.	CO 3 CO 4	12	12
Files: Introduction to files in C/C++; opening; closing and updating binary and sequential files. Advanced topics: Operations on bits; Register variable; Pre-processors in C.	CO 2 CO 3 CO 4	9	9

Recommended Books:

1. E. Balagurusamy: Programming with ANSI C, *Tata McGraw Hill*
2. Byron Gotfried : Programming with C
3. Robert Lafore : Programming with C
4. Herbert Schildt : Teach yourself C

MATH 1101: Differential Calculus and Co-Ordinate Geometry	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- Introduce the students to Sketch the plane given by a linear equation
- Enhance their analyzing and problem-solving skills the distance between points, planes, and lines in space also recognize and write equations for different surfaces and different theorems

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Understand and appropriately use the technical vocabulary of the topics covered such as Limit; Continuity and differentiability
CO 2	Understand and appropriately use the Leibnitz's theorem; Rolle's theorem; Mean value theorems; Taylor's theorem
CO 3	Perform the use of co-ordinates axes and its uses geometrically
CO 4	Understand and appropriately use the equations of parabola, ellipse and hyperbola in cartesian and polar co-ordinates

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√	√										
CO2	√	√	√									
CO3		√	√									
CO4	√	√										

Course Content:

Course Topics	CO	Contact Hours
Differential Calculus: Limit; Continuity and differentiability; Differentiation of explicit and implicit function and parametric equations; Significance of derivatives; Differentials; Successive differentiation of various types of functions;	CO 1	9
Leibnitz's theorem; Rolle's theorem; Mean value theorems; Taylor's theorem in finite and infinite forms; Maclaurin's theorem in finite and infinite forms; Langrange's form of remainders; Cauehy's form of remainder; Expansion of functions by differentiation and integration; Partial differentiation; Euler's theorem; Tangent and Normal; Maxima and Minima; Points of inflection and their applications; Evaluation of indeterminate forms by L'Hospitals rule.	CO 2	12
Co-ordinate Geometry: Transformation of co-ordinates axes and its uses; Equation of conics and its reduction to standard forms; Pair of straight lines; Homogeneous equations of second degree; <u>Angle between a pair of straight lines;</u>	CO 3	12
Pair of lines joining the origin to the point of intersection of two given curves; circles; System of circles; Orthogonal circles; Radical axis, radical center, properties of radical axes; Coaxial circles and limiting points; Equations of parabola, ellipse and hyperbola in cartesian and polar co-ordinates.	CO 4	9

Recommended Books:

1. Das and Mukharjee : Differential calculus
2. J.Edwards : Differential calculus
- 3.H.H.Ashwith : Analytic geometry of conic systems
4. M.L. Khanna : Solid Geometry

PHY 1101: Physics	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- Give students an in-depth understanding structure of Matter, Atomic Physics
- Introduce the fundamentals of Waves and Oscillations and theories of Light.
- Provide hands-on use Interference, Diffraction and Polarization

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Bridge the fundamental concepts of Structure of matter and atomic model
CO 2	Learn and real time application of simple harmonic motion, composition of simple harmonic motions
CO 3	Identify categories of light sources, Huygens wave theory, Huygen's principle and construction, Superposition of light waves.
CO 4	Recognize various Diffraction and also by single and double slit and polarization.

CO No	Program Learning Outcomes (PLO)											
	PLO1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√											
CO3	√	√										
CO4	√	√										

Course Content:

Course Topics	CO	Contact Hours
Structure of Matter: Structure of matter. Different types of bonds in solids: metallic, van der Waals'; Covalent and Ionic bond, Packing in solids: Inter atomic distances and forces of equilibrium, X-ray diffraction, Bragg's law. Atomic Physics: Atom models: Thomson atom model, Rutherford atom model, Rutherford scattering formula, Electron orbits, Bohr atom model; Energy levels and spectra; Particle properties of waves.	CO 1	9
Waves and Oscillations: Oscillations: Simple harmonic motion, Composition of simple harmonic motions and Lissajous' figures, Damped and Forced oscillations; Resonance; Waves: Wave properties of particle: de Broglie	CO 2	12

waves, Group velocity, Phase velocity; Travelling and Standing waves; Energy calculation of Traveling and Standing waves; Intensity of waves; Beats; Doppler effect.		
Theories of Light: Wave theory: Huygens wave theory; Huygen's principle and construction; Superposition of light waves; Electromagnetic theory; Particle theory: Newton's corpuscular theory, Quantum theory of light. Interference: Introduction; Conditions of Interference; Young's double slit experiment; Fresnel's biprism; Thin film Interference; Interference due to multiple reflection.	CO 3	12
Diffraction: Fresnel's and Fraunhofer diffraction; Diffraction by single and double slit; Diffraction gratings. Polarization: Introduction; Methods of producing polarized light; Polarization by reflection and refraction; Polarization by double refraction; Construction of Nicol prism; Production and analysis of polarized light; Optical activity; Optics of crystals; Polarimeters.	CO 4	9

Recommended Books:

1. B.L. Theraja : Modern Physics
2. Kenneth Krane : Modern Physics
3. Brijlal : A textbook of Optics
4. by J Peatross & Michel ware : Physics of light and optics
5. S.P.Puri : Waves & Oscillations

HUM 1101: Communicative English	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
HUM 1102: Communicative English Lab	
Credits: 0.75	Contact Hours: 0L+3/2P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- Heighten their awareness of correct usage of English grammar in writing and speaking
- Improve their speaking ability in English both in terms of fluency and comprehensibility
- Give oral presentations and receive feedback on their performance
- Increase their reading speed and comprehension of academic articles
- Improve their reading fluency skills through extensive reading
- Enlarge their vocabulary by keeping a vocabulary journal
- Strengthen their ability to write academic papers, essays and summaries using the process approach.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Review the grammatical forms of English and the use of these forms in
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	specific communicative contexts, which include: class activities, homework assignments, reading of texts and writing
CO 2	Develop their ability as critical readers and writers
CO 3	Produce a short research paper using the drafting process
CO 4	Free hands writing in different form like essay, paragraph, report etc.
CO 5	Attain and enhance competence in the four modes of literacy: writing, speaking, reading and listening

CO No	Program Learning Outcomes (PLO)											
	PLO1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1										√		√
CO2				√					√	√		
CO3										√		√
CO4									√	√		√
CO5									√	√		√

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction: Parts of Speech (Changing); Sentence making rules; Different types of sentences; Tense; Right forms of verbs; Verb conjugation. Transformation of sentences: Different types of sentences; Converting degree; Appropriate preposition; Idioms & Phrase; Voice; Narration; Correction of sentences; WH Question; Completing sentences; Synonyms and Antonyms.	CO 1	12	3
English in Communication: Introductions one-self; Conversation & Dialogue; Group Discussion; Notations & Functions. Paragraph Writing: Amplification of Ideas; Structure and Classifications; Topic Sentence; Topic Developers; Topic Termination; Open-ended Paragraph; Close-ended Paragraph.	CO 2 CO 3	9	9
Essay Writing: A Hints given Essay; Without hints Essay; Free hand Essay writing on current Issues. Reading Comprehension: Precise writing; Vocabulary building; Synonym & Antonym; Use of words in different Parts of Speech; True-	CO 4	12	3

False (Yes/No/ Not given); Fill in the Blanks; Tree Chart, Flow Chart. Business Writing: Agenda; Notice; Memo; Meeting Minutes; Quotation; Tender; Letter Writing: Job application, Resume; Formal & Informal letter; Letter to Newspaper; Report writing: Technical Report, Lab Report, Newspaper Report.			
Developing Reading Skill: Strategies of reading skimming, scanning, predicting, inferencing; Practicing comprehension from literary and non literary texts. Developing Writing Skill: Sentence variety; Generating sentences; Clarity and correctness of sentences; Linking sentences for paragraphs; Writing paragraphs, essays, reports, formal and informal letters. Developing Listening Skill: Listening to recorded texts, class lectures and learning to take notes. Developing Speaking Skill: Oral skills including communicative expressions for personal identification, life at home; Giving advice and opinion; Instruction and directions, requests, complains apologies; Describing people and places; Narrating events.	CO 5	9	6

Recommended Books:

1. Cambridge ESOL : Cambridge IELTS 1-9 Students Book with Answers
2. Philpot, Sarah, Curnick, Lesley : New Headway Academic Skills: Reading, Writing and Study Skill, Oxford University Press ELT
3. Martin Hewings : Cambridge Academic English, Cambridge University Press
4. Collins Education : Collins English Skills
5. John Langan : College Writing Skills

CSE 1150: Viva-Voce	
Credits: 0.75	Contact Hours: 0L+0P Hrs/Week

The viva-voce is designed to attain comprehensive knowledge about the whole semester along with that one can improve his/her attitude.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	The viva-voce is designed to attain comprehensive knowledge about the whole semester along with improvement of his/her communication and attitude.
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CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1										✓		✓

YEAR-1 TERM-2 B. Sc. ENGINEERING COURSES

ME 1200: Engineering Drawing	
Credits: 1.0	Contact Hours: 0L+2P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- Learn different types of object drawing.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Understand the importance of BIS and ISO Standards in Engineering Drafting
CO 2	Graphically construct and understand the importance of mathematical curves in Engineering applications
CO 3	Visualize geometrical solids in 3D space through exercises in Orthographic Projections, Orthographic ,Isometric and Perspective views of objects
CO 4	Develop the surfaces of geometrical solids

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	✓											
CO2	✓											
CO3	✓			✓								
CO4		✓		✓	✓							

Course Content:

Course Topics	CO	Contact Hours
Basic Scale drawing	CO 1	3
Isometric views , Orthographic view Missing line, Sectional view, Auxiliary view	CO 2 CO 3	6
Project on Engineering Drawing and CAD using Auto CAD.	CO 4	5

Recommended Books:

1. Engineering Drawing & Design : Cencil Jensen, Jay D. Helsel, Dennis R. Short, Seventh Edition , Tata McGrawHill 2012

CSE 1200: Structured Programming Language Sessional II	
Credits: 0.75	Contact Hours: 0L+3/2P Hrs/Week

Advanced sessional works based on Structured Programming Language.

CSE 1201: Object Oriented Programming	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 1202: Object Oriented Programming Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- General principles of programming in object oriented frameworks to enhance transferable skills, such as programming,
- Designing, and problem-solving skills.
- Introduces object oriented concepts and develops OOP programs which provides solutions to real world object oriented problems

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Identify issues related to the definition, creation and usage of classes and objects.
CO 2	Use the principles of inheritance and polymorphism and analyses problems for design of abstract classes and interfaces.
CO 3	Apply programming skills and examine key issues to the design of object oriented software.
CO 4	Choose software tool, perform and demonstrate skills, and write report to design, build, and test realistic object oriented programs.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√	√		√							
CO3		√	√		√							
CO4		√	√	√					√		√	

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Fundamentals of OOP: Introduction to Object Oriented Programming; Principles of Object Oriented Design; Advantages of OOP over structured programming. Classes and Objects: Structure of classless; Public, private and protected members; Array of objects; Nested member class and their object; Pointer objects and pointer members; Static class member and static class; Friend function.	CO 1	14	8
Constructor and destructors: Default constructor; Argumented constructor; Copy constructor; Dynamic constructor; Constructor function for derived class and their order of execution; Destructor. Inheritance: Inheritance basics; Single inheritance vs. multiple inheritance; Mode of inheritance; Virtual inheritance. Polymorphism: Abstract classes; run-time and compile time polymorphism; Virtual function overloading; Run Time Type Identification (RTTI); Unary and binary operator overloading.	CO 2 CO 3	20	20
Advanced Topics: Exception Handling; Template functions and classes, Namespace; Standard Template Library, Multi-threaded programming.	CO 3 CO 4	8	14

Recommended Books:

1. Matt Weisfeld : The Object-Oriented Thought Process.
2. Brett D. McLaughlin, Gary Pollice, Dave West: Head First Object-Oriented Analysis and Design.
3. Martin Fowler : Refactoring: Improving the Design of Existing Code.
4. Herbert Schildt : Teach Yourself C++.

CSE 1203: Discrete Mathematics**Credits: 3.0****Contact Hours: 3L+0P Hrs/Week****Course Objective:** At the end of the course a student is expected to:

- Address mathematical reasoning, combinatorial analysis, algorithmic thinking, and discrete structures
- To designing, and problem-solving skills.

- To emphasize on mathematical definitions and proofs as well as methods of application.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Understand inductive proofs
CO 2	Understand and apply predicate logic to solve problems
CO 3	Determine the truth of statements
CO 4	State and apply facts, and notation of commonly used discrete math constructs

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√										
CO3		√										
CO4	√	√	√									

Course Content:

Course Topics	CO	Contact Hours
Logic and Proofs: Propositional Logic; Propositional Equivalences; Predicates and Quantifiers; Nested Quantifiers; Rules of Inference; Introduction to Proofs; Proof Methods and Strategy.	CO 1	9
Sets and Functions: Sets; Set Representations; Subsets; Power set; Cartesian Products; Set Identities; Operations on sets; Laws of set operation; Properties and laws of various set numbers; Functions; One-to-One and Onto functions; Inverse functions; Compositions of functions; Floor and ceiling functions.	CO 2	9
Induction and Recursion: Mathematical Induction; Strong Induction and Well-Ordering; Recursive Definitions and Structural Induction; Recursive Algorithms. Counting: Basics of counting; Pigeonhole Principle; Permutations and Combinations; Advanced Counting Techniques. Relations: Introduction to Relations; Functions as Relations; Properties of Relations; Combining Relations; Representing Relations: Matrices and Digraphs; Closures of Relations.	CO 3	12
Graph: Graphs and Graph models; Graph terminology and special types of Graphs; Representing Graphs; Isomorphism of Graphs; Connectivity; Graph Algorithms; Planner Graph; Graph Colouring.	CO 4	12

Trees: Introduction to trees; Properties of trees; Different types of trees; Application of trees; Tree traversal algorithms; Infix, Prefix and Postfix Notations ; Spanning trees.		
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Recommended Books:

1. Kenneth H. Rosen: Discrete Mathematics and its Applications, Tata McGraw-Hill.
2. Donald F. Stanat and David F. McAllister: Discrete Mathematics in Computer Science, Prentice Hall College Div.
3. C. L. Liu: Elements of Discrete Mathematics, Tata McGraw-Hill.
4. Norman L. Biggs: Discrete Mathematics, Oxford University Press.
5. Thomas Koshy: Discrete Mathematics with Applications, Elsevier.

EEE 1201: Basic Electrical Engineering	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
EEE 1202: Basic Electrical Engineering Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- Impart a basic knowledge of electrical quantities such as current, voltage, power, energy and
- Provide working knowledge for the analysis of basic DC and AC circuits used in electrical and electronic devices
- To explain the working principle, construction, applications of DC machines, AC machines & measuring instruments.
- Highlight the importance of transformers in transmission and distribution of electric power.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn the behavior of any electrical and magnetic circuits.
CO 2	Formulate and solve complex AC, Dc circuits.
CO 3	Identify the type of electrical machine used for that particular application.
CO 4	Realize the requirement of transformers in transmission and distribution of electric power and other applications.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√										
CO3	√											
CO4	√	√										

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Electric current and Ohm's law: Fundamental	CO 1	14	14

electrical concepts and measuring units; Electron drift velocity; Charge velocity and velocity of field propagation; Electric potential; Conductance and conductivity; Variation of resistivity with temperature; Ohm's law; Resistors in series and parallel; Voltage divider rule; Short and open circuits; Kirchoff's current law (KCL) and Kirchoff's voltage law (KVL); Problems solved by these law.			
Networks Theorems: Independent and dependent sources; Superposition theorem; Millman's theorem; Reciprocity theorem; Thevenin's theorem; Norton's theorem; Maximum power transfer theorem; Mesh and Node circuit analysis; Reduction of complicated networks; T and p-section network. Alternating Current: General AC theory; AC power; Average and RMS value of AC voltage and current; Use of complex quantities in AC circuits; Resonant circuits; Q-value and bandwidth.	CO 2 CO 3	14	14
DC Generator: Principles; construction; classification; armature winding; voltage build up; armature reactions and commutation; losses of efficiency; parallel operation of DC generators. DC Motor: Operation; types; losses and efficiency; starting; separately excited DC motor; permanent magnet DC motor; speed-torque characteristics; methods of starting and speed control; applications of DC motors. Transformer: Principle; construction; vector diagrams and voltage regulations; equivalent circuits; losses and efficiency.	CO 3 CO 4	14	14

Recommended Books:

1. Charles K. Alexander, Matthew N.O. Sadiku : Fundamentals of Electric Circuits.
2. Robert L. Boylestad : Introductory Circuit Analysis
3. Rosenblatt, FRIEDMAN : Direct and Alternating Current Machinery
4. B.L. Theraja : Textbook of Electrical Technology
5. A. K. Sawhney : A Course in Electrical

MATH 1201: Integral Calculus and Ordinary Differential Equations and Series Solutions

Credits: 3.0

Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- Identify a system of linear equations (or linear system) and describe its solution set
- Relate the solution set of a consistent inhomogeneous linear system to the solution set of its associated homogeneous equation

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn the echelon forms in finding the solution sets of linear systems
CO 2	Learn whether sets of vectors are linearly independent or dependent
CO 3	Compute the inner product of vectors, lengths of vectors, and determine if vectors are orthogonal
CO 4	Compute standard operations on vectors in two-dimensional space and three dimensional space

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√										
CO3		√	√									
CO4		√	√									

Course Content:

Course Topics	CO	Contact Hours
Integral Calculus: Anti-derivative and Techniques of Indefinite integrals; Definite integrals: Geometric interpretation of definite integrals; Fundamental theorem of calculus I and II; General properties;	CO 1	9
Evaluation of definite integrals. Reduction formulas; Applications of definite integrals to find area and arc-length of 2-curves; Solid revolutions; Volume and surface area of hollow and solids bodies; Improper integrals; Convergence of improper integrals; Beta and gamma functions, their properties and applications to solve integrals	CO 2	9
Solution of homogeneous linear differential equations; Solution of higher order differential equations when the dependent of independent variables are absent; Solution of differential equation with constant coefficients by operator method; Differential equations with variable coefficients.	CO 3	12
Series Solution: Solution of differential equations in series by the method of Frobenius; Bessel's functions, Legendre's polynomials and their properties.	CO 4	12

Recommended Books:

1. Das and Mukharjee : Integral Calculus
2. A.K.Hazra : Integral Calculus
3. S L Ross : Differential Equations
4. Frank Ayres : Differential Equations

HUM 1201: Economics	
Credits: 2.0	Contact Hours: 2L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- Explain the behavior of buyers and sellers in the market using basic economic theories
- Explain the role of government in influencing buyer and seller behavior in the market
- Recognize and analyze common economic issues which relate to individual markets and the aggregate economy

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn basic economic theories that explain economic outcomes of the aggregate economy
CO 2	Outline the implications of various economic policies on individuals and on the economy
CO 3	Justify how your individual decisions and actions as a member of society affect the economy locally, nationally and internationally.
CO 4	Understand basic economic principles that influence global trading and the challenges relating to globalization
CO 5	Understand, analyze and describe the budget and policy

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√					√		√				√
CO2						√	√	√			√	√
CO3						√			√			√
CO4							√					√
CO5						√	√	√			√	

Course Content:

Course Topics	CO	Contact Hours
Basic Concepts of Economics: Definition and subject matter of Economics; Microeconomics vs macroeconomics; Law of Economics; Central economic problems of every society; Different economic systems; Economics and Engineering.	CO 1	9
Theory of Demand, Supply and Consumer Behavior: Law of Demand; Demand schedule and demand curve; Supply law, Supply schedule and supply curve; Shift in demand and supply; Equilibrium in the market; Elasticity of demand and supply Production and Costs and Theory of the Firm: Meaning of production; Factors of production; Concepts of total, average and marginal costs, fixed and variable costs.	CO 2	9
Basic Concepts of Macroeconomics: Growth; Unemployment; Inflation; Philips Curve, Business cycle; Circular flow of economics; Two, three and four sector economics.	CO 3	9
National Income accounting and determination: Concepts of GNP, GDP and national income; Methods of national income accounting; Problems of national income accounting; Keynesian model of national income determination; The multiplier; Effect of fiscal policy in the Keynesian model.	CO 4	9
Budgets of Bangladesh: The revenue at the capital budget; Income, expenditure of the government; direct and indirect taxes.	CO 5	6

Recommended Books:

1. K. K. Dewett: Modern Economic Theory
2. H.L. Ahuja: Advanced Economic Theory
3. A. Asimakopulos: An Introduction To Economic Theory
4. A. Koutsoyiannis: Modern Microeconomics

CSE 1250: Viva-Voce	
Credits: 0.75	Contact Hours: 0L+0P Hrs/Week

The viva-voce is designed to attain comprehensive knowledge about the whole semester along with that one can improve his/her attitude.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	The viva-voce is designed to attain comprehensive knowledge about the whole semester along with improvement of his/her communication and
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	attitude.
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CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1										✓		✓

YEAR-2 TERM-1 B. Sc. ENGINEERING COURSES

CSE 2101: Data Structures	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 2102: Data Structures Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- To impart the basic concepts of data structures and algorithms
- To understand concepts about searching and sorting techniques
- To Understand basic concepts about stacks, queues, lists, trees and graphs
- To understanding about writing algorithms and step by step approach in solving problems with the help of fundamental data structures

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn various data structures and its operations.
CO 2	Analyze and implement searching, insertion, deletion, traversing mechanism etc. on various data structures.
CO 3	Apply data structures concepts in various domains like DBMS, compiler construction etc.
CO 4	Apply linear and non-linear data structures like stacks, queues , linked list, tree, graph etc.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	✓											
CO2		✓	✓									
CO3		✓	✓		✓							
CO4		✓	✓		✓							

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction: Concepts and Examples of Elementary Data Objects; Necessity of Structured Data; Types of Data Structure; Ideas on Linear and Nonlinear Data Structure.	CO 1	6	6
Linear Array: Linear Array& its representation in memory; Traversing LA; Insertion & Deletion in LA; Bubble Sort, Linear Search and binary Search; Multidimensional Array and its representation in memory; Algebra of matrices; Sparse matrices; Stack: Stack representation and applications; PUSH and POP operation on stack; Polish Notation, reverse polish notation; Evaluation of a postfix expression; Transforming infix expression into postfix expression.	CO 2	6	6
Queue: Its representation; Insertion & deletion in Queue; Priority Queues; Recursion [Factorial function, Fibonacci sequence, Ackermann function, Towers of Hanoi]. Linked List: Linked list and its representation in memory; Traversing, Searching, Insertion and Deletion operation on Linked list; Circular List; Header linked lists; Two way lists.	CO 3	9	9
Complexity: Algorithm and flow chart; Complexity of algorithms; Rate of growth; Big O notation; Complexity of Linear Search; Binary search and Bubble sort algorithm. Sorting: Insertion sort, selection sort, quick sort, merge sort; Searching and data modification; Hash function; Collision resolution, Chaining.	CO 4	9	9
Graphs: Introduction, Definitions and terminology, Graph representations, Traversals, Shortest path and transitive closure, Activity networks, Topological sort and critical paths, Graph Algorithms- Breadth First Search, Depth First Search, Minimum Spanning Tree, Kruskal and Prims Algorithm. Tree: Tree terminology; Representation of binary trees in memory; Traversing binary tree; Binary search tree; Insertion and deletion on binary search tree; Insertion and deletion on heap; Heap sort; B trees; General tree.	CO 5	12	12

Recommended Books:

1. Seymour Lipschetz : Data Structure
2. Y. Langsam, Augenstein, A. M.Tanenbaum: Data Structures Using C and C++
3. Edward M. Reingold : Data structures
4. Robert Sedgwick : Algorithms in C
5. Niklaus wirth : Algorithms and Data Structures Program

CSE 2103: Design Pattern and Java Programming	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 2104: Design Pattern and Java Programming Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- General principles of programming in object oriented frameworks to enhance transferable skills, such as programming,
- Designing, and problem-solving skills.
- introduces object oriented concepts and develops OOP programs which provides solutions to real world object oriented problems

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Identify issues related to the definition, creation and usage of classes and objects.
CO 2	Apply programming skills and examine key issues to the design of object oriented software.
CO 3	Choose software tool, perform and demonstrate skills, and write report to design, build, and test realistic object oriented programs.
CO 4	Understand and use java program in networking

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√		√									
CO2		√	√		√							
CO3				√	√							
CO4	√				√	√						

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction: Java Programming Environment, Fundamental Programming Structures in Java, Object and Classes, Inheritance, Interfaces, Exceptions, Streams and I/O Programming, Polymorphism, Multi-threaded Programming. Design Tools: Introduction to UML, Identification of Scenarios and UML Use Case Diagrams, UML Interaction Diagram, Scenario Elements in terms of CRC Models.	CO 1	9	9
Design Patterns: Introduction, Creational Patterns - Abstract Factory, Builder, Factory Method, Prototype, Singleton; Structural Patterns - Adapter, Bridge, Composite, Decoder, Proxy, Facade, Flyweight; Behavioral Patterns- Chain, Command, Interpreter, Iterator, Mediator, Memento, Observer, State, Strategy, Visitor, Template Method.	CO 2	12	12
Java GUI Programming: Creating Windows, Events Handling, User Interface Components with Swing, Applets, Java Server Pages, Java Servlets, Java Beans, JDBC Programming.	CO 3	9	9
Java Network Programming: Client-Server Model, Socket Programming, Distributed Computing, RMI-based Application.	CO 4	12	12

Recommended Books:

1. Rajkumar Buyya, S. T. Selvi and X. Chu: Object –Oriented Programming with Java: Essentials and Applications, McGraw Hill
2. Deitel & Deitel: Java How To Program, Prentice Hall
3. McConnell: Code Complete: a practical handbook of software construction.
4. E. Gamma (aka Gang Of Four): Design Patterns: Elements of Reusable OO Software.
5. Cay S. Horstmann: Object-Oriented Design and Patterns

EEE 2101: Electronic Devices and Circuits	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
EEE 2102: Electronic Devices and Circuits Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- Impart a basic knowledge of electrical quantities such as Diodes, Energy bands in solids, Valence and conduction band
- Provide working knowledge for the analysis of basic P-N junction diodes and their V-I characteristics; Zener diode; Tunnel diode
- To explain the working principle FET, MOSFET, Classification of MOSFET, CMOS.
- To explain the working principle TTL, ECL, IIL and CMOS

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn the behavior of any Diodes, Energy bands in solids, Valence and conduction band
CO 2	Formulate and solve complex basic P-N junction diodes and their V-I characteristics; Zener diode; Tunnel diode
CO 3	Identify and understand the type of FET, MOSFET, Classification of MOSFET, CMOS.
CO 4	Realize the requirement for TTL, ECL, IIL and CMOS.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√	√									
CO3	√	√										
CO4	√			√								

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Semiconductor Diodes: Energy bands in solids; Valence and conduction band; Classification of solids in terms of energy bands; Work function, Electron emission; Intrinsic and Extrinsic semiconductors, n-type and p-type semiconductors, P-N junction diodes and their V-I characteristics; Zener diode; Tunnel diode; Varactor diode; Thermistor; Photodiode and LDR; Transition and Diffusion capacity.	CO 1	9	9
Diode Circuits: Ideal rectifier concept; Half wave and Full wave rectifiers; High Precision rectifiers; Filters; Voltage regulators; Voltage doubler;	CO 2	12	12

Clippers; Clampers. Bipolar Junction Transistors: Working principle of PNP and NPN transistor; Transistor as an amplifier, Common-base, Common-emitter and Common Collector Configurations; Input and output characteristics of CB, CE, and CC transistor connections; Load line analysis; Operating point; Performance of transistor operation; Cutoff and Saturation points.			
Field-Effect Transistors (FET):Construction and classification; Principle of operation; Characteristic curves; Channel ohmic and pinch-off region; Characteristic parameters of the FET; Effect of temperature on FET, MOSFET, Classification of MOSFET, CMOS. Operational Amplifier: Difference amplifier CMRR, Ideal operational amplifier; Inverting amplifier, Non-inverting amplifier; Summing Amplifier, Integrator, Differentiator, Linear and non-linear applications of operational amplifier; Comparator and Converter; 741 monostable and bistable multivibrator.	CO 3	12	12
Oscillators: Positive feedback; Condition of oscillation; RC phase shift oscillator; Wein bridge oscillator; Resonant circuit oscillators; Crystal oscillator and Waveform generators. Others:Diode logic gates; Transistor switches; Transistor gates; MOS gates; Logic Families: TTL, ECL, IIL and CMOS logic with operation details; Propagation delay, product and noise immunity; Open collector and high impedance gates.	CO 4	9	9

Recommended Books:

1. Jacob Millman and Christos C. Halkias: Electronic Devices and Circuits.
2. Albert D. Helfrick and William David Cooper: Modern Electronics Instrumentation and Measurement Techniques.
3. Albert Paul Malvino: Electronic Principles.

MATH 2101: Vector, Matrices and Linear Algebra

Credits: 3.0

Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- Identify a system of fundamental laws, product of vectors, geometrical and physical interpretations
- Relate the solution set of a consistent LU decomposition, QR decomposition, Cholesky decomposition, Jordan Decomposition, Spectral decomposition

- Identify linear transformation, kernel, image, nullity, rank. Matrix representation of linear operator

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn fundamental laws, product of vectors, geometrical and physical interpretations
CO 2	Learn and understand solution set of a consistent LU decomposition, QR decomposition, Cholesky decomposition, Jordan Decomposition, Spectral decomposition
CO 3	Compute the Singular value decomposition, Applications. System of linear equations
CO 4	Apply standard operations on kernel, image, nullity, rank. Matrix representation of linear operator

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√										
CO3		√	√									
CO4		√	√									

Course Content:

Course Topics	CO	Contact Hours
Vector Analysis: Vectors and scalars: definitions and fundamental laws, product of vectors, geometrical and physical interpretations, reciprocal vectors. Vector Geometry: equation of planes, straight lines and spheres.	CO 1	9
Vector differentiation, Vector differential operators, gradient, divergence, curl and their physical interpretations. Vector integration, Green's theorem, Gauss' theorem and Stokes' theorem and their applications. Curvilinear co ordinates.	CO 2	12
Matrices and Linear Algebra: Definitions and properties of different types of Matrices, operations for matrices and related theorems. Transpose of a matrix and inverse matrix, rank of a matrix. Matrix Decompositions: Introduction, LU decomposition, QR decomposition, Cholesky decomposition, Jordan Decomposition, Spectral decomposition, Singular value decomposition, Applications.	CO 3	12
System of linear equations: Echelon, normal and canonical forms of matrices. Consistency and solutions of homogeneous and non-homogeneous systems of linear equations by matrix	CO 4	9

methods. Vector spaces and subspaces, sum and direct sum, linear dependence and independence, basis and dimension. Linear transformation, kernel, image, nullity, rank. Matrix representation of linear operator, change of basis, similarity, matrices and linear mapping. Characteristic roots and characteristic vectors of linear transformations and applications..		
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Recommended Books:

1. M.R.Spigel : Vector and Tensor Analysis
2. Frank Ayres : Matrices
3. M. R. Spiegel : Linear Algebra
4. Seymour Lipschutz, Marc Lipson : Linear Algebra, Schaum's Outline Series, McGraw-Hill

STAT 2101: Elementary Statistics and Probability	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- Identify a system of fundamental Variables and Attributes. Collections and presentation of statistical
- Relate the solution set of Sample Space, Events, Union and Intersection of Events
- Identify linear continuous random variables. Density and distributional functions

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learnto use of system of Variables and Attributes. Collections and presentation of statistical
CO 2	Solve set of Sample Space, Events, Union and Intersection of Events
CO 3	Compute the random variables,density and distribution functions
CO 4	Compute and apply standard operations Multinomial, Cauchy and Wibil distribution.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√										
CO3		√	√									
CO4		√	√									

Course Content:

Course Topics	CO	Contact Hours
Statistics: Meaning and Scope, Variables and Attributes. Collections and presentation of statistical data. Frequency distribution and Graphical Representation. Analysis of Statistical Data: Location, Dispersion and their measures. Skewness, Kurtosis and their measures; Moment and cumulants.	CO 1	9
Elements of Probability: Sample Space, Events, Union and Intersection of Events. Probability of Events. Loss of probability. Frequency limit and probabilities. Addition law of probability. Application to Occupancy problems, Bose-Einstein statistics, Fermi-Dirac statistics, Conditional probabilities. Bayes probability, Chebysev's Inequality.	CO 2	9
Random Variables and Probability Distribution: Basic concepts. Discrete and continuous random variables. Density and distributional functions. Mathematical expectation and variance. conditional expectation and conditional variance,	CO 3	12
Expected values and variance of the density distributions. Moments and cumulants generating functions, Characteristics function. Study of Binomial, Poisson, Normal, Geometric, Negative binomial, Hypergeometric, Multinomial, Cauchy and Wibul distribution.	CO 4	12

Recommended Books:

1. S.C. Gupta and V.K. Kapoor : Fundamentals of Mathematical Statistics
2. R.N. Shill & S.C. Debnath : An introduction to the theory of Statistics
3. M.G. Mostafa : Methods of Statistics
4. Dr Manindra Kumar Roy : An Introduction to the theory of Probability

CSE 2150: Viva-Voce	
Credits: 0.75	Contact Hours: 0L+0P Hrs/Week

The viva-voce is designed to attain comprehensive knowledge about the whole semester along with that one can improve his/her attitude.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	The viva-voce is designed to attain comprehensive knowledge about the whole semester along with improvement of his/her communication and attitude.
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CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1										√		√

YEAR-2 TERM-2 B. Sc. ENGINEERING COURSES

CSE 2200: Analytical Programming Sessional	
Credits: 1.5	Contact Hours: 0L+3P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- The advance learning of Programming Language
- Solve challenging problems and prepare students to compete programming contest through various Programming Language.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Identify different issues of advance programming problems
CO 2	Apply and solve challenging problems and prepare students to compete programming contests.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√	√	√	√			√					
CO2		√	√	√	√				√			√

Course Content:

Course Topics	CO	Contact Hours
Student will solve at least 30 problems (ACM-ICPC, NCP, ICFP, UVA like) using C, C++ or Java. Among them at least three problems should be submitted from Geometry, Mathematics, String Processing, Tree, Graph and Sorting Techniques.	CO 1 CO 2	42

Recommended Books:

1. Rajkumar Buyya, S. T. Selvi and X. Chu: Object –Oriented Programming with Java: Essentials and Applications, McGraw Hill
2. Deitel & Deitel: Java How To Program, Prentice Hall

CSE 2201: Algorithms	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 2202: Algorithms Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- To study efficient algorithms and effective algorithm design techniques
- Analysis of algorithms, Methods for the design of efficient algorithms

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn various methods for solving a specific problem
CO 2	Learn techniques for designing efficient algorithms
CO 3	Evaluate the performances of algorithms
CO 4	Apply various algorithms to solve different problems

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√	√									
CO3		√	√	√								
CO4		√	√		√							

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Asymptotic notations: Complexity analysis of algorithms, worst case, best case and average case.	CO 1	3	3
Sorting algorithms: Divide and Conquer approach, Merge Sort and Quick Sort Algorithm, complexity analysis, worst and average case analysis, Heap Construction Algorithm, Heap sort, Application of Heap: Priority Queue, Decision tree model and (worst case) lower bound on sorting, Sorting in linear time - radix sort, bucket sort, counting sort, etc.	CO 2	9	9
Graph algorithms: Representation of Graphs,	CO 3	9	12

Breadth First Search, Depth First Search, Minimum Spanning Tree, Kruskal and Prims Algorithm. Shortest Path algorithms: Dijkstra's Algorithm, Bellman-Ford Algorithm. Floyd Warshall Algorithm			
Searching algorithms: Binary search trees, balanced binary search trees, AVL trees and redblacktrees, B-trees, skip lists, hashing. Priority queues, heaps, Interval trees. Dynamic Programming: Longest Common Subsequence (LCS), Matrix Chain Multiplication (MCM). Greedy Algorithm: Greedy Algorithm, Activity Selection Problem, Huffman Codes and its application, Knapsack problem, Traveling Salesperson Problem.	CO 4	9	9
Recurrences and Backtracking: Recurrences, NP-Hard and NP-Complete Problems, Backtracking, n-Queen Problem, Branch and Bounds. Reducibility between problems and NP-completeness: Lower bound theory, Discussion of different NP-complete problems like satisfiability, clique, vertex cover, independent set, Hamiltonian cycle, TSP, knapsack, set cover, bin packing, etc. Computational Geometry, Line Segment Properties, Convex Hull, Graham Scan Algorithm of Convex Hull.	CO 3 CO 4	12	12

Recommended Books:

1. Cormen : Introduction to Algorithms
2. Horowitz, Shanny : Computer Algorithms
3. D. E. Knuth : The art of Computer Programming
4. M. Allen : Data Structure and Algorithm analysis in C++.

CSE 2203: Theory of Computation	
Credits: 2.0	Contact Hours: 2L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study finite representation objects and basic mathematical models of computation
- analysis of finite automata and regular languages, pushdown automata and context free languages

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn finite representation of infinite objects and basic mathematical models of computation
CO 2	Understand and apply regular languages, pushdown automata and context free languages
CO 3	Evaluate the performances Turing machines. Church's Thesis. Partial recursive functions
CO 4	Learn time complexity and NP-completeness, probabilistic computation, interactive proof systems.
CO 5	Apply the models in real life project.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√	√		√							
CO3		√	√	√								
CO4		√	√		√							
CO5		√	√					√	√			

Course Content:

Course Topics	CO	Contact Hours
Finite State Machine: Fundamental of finite state machine, state equivalence and minimization of machine, incompletely specified machine and minimal machine, merger graph and compatibility graph, finite memory and definite memory machine, information lossless machine and inverse machine.	CO 1	3
Finite Automata: Introduction to finite automata, structural representations, automata and complexity, An informal picture of finite automata, Deterministic finite automata, nondeterministic finite automata, an application (i.e. text search or other), finite automata with Epsilon-Transitions.	CO 2	3
Regular Expressions and Languages: Regular expressions, finite automata and regular expression, application of regular expressions, algebraic laws for regular expressions; Closure properties of regular language, Decision properties of regular languages, equivalence and minimization of automata.	CO 3	6
Context-Free Grammar and Languages: Context-free grammars, parse trees, application of context-free grammars,	CO 4	6

ambiguity in grammars and languages, Normal form for contextfree grammars, the pumping lemma for context-free languages, closure properties of contextfree languages, decision properties of CFL's.		
Pushdown Automata: Definition of the pushdown automata, the languages of a PDA, equivalence of PDA's and CFG's, deterministic pushdown automata. Introduction to Turing Machines: The turing machine, programming techniques for turing machines, Extensions to the basic turing machine, restricted turing machines, turing machines and computers.	CO 2 CO 3 CO 5	3

Recommended Books:

1. John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman : Introduction to Automata Theory, Languages, and Computation, *Pearson/Addison Wesley*
2. Shyamalendu Kandar : Express Learning: Automata Theory and Formal Languages, *Pearson Education India*
3. Zvi Kohavi : Switching and Finite Automata

CSE 2205: Digital Systems	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 2206: Digital Systems Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- To study different digital logics, its functions and its implementation with different hardware.
- Analysis different combinational and sequential circuits and their application will be studied in details.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Ability to understand basics of different logic, functions and application of digital logic
CO 2	Ability to implement different logic functions using different hardware
CO 3	Ability to implement different Boolean functions using different hardware.

CO No	Program Learning Outcomes (PLO)
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	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√	√	√								
CO3		√	√	√								

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Information and digital Systems: Introduction to digital systems, Number Systems, weighted and non-weighted codes, error detection code, Binary addition and subtraction, 2's complement methods.	CO 1	9	3
Boolean Algebra and Combinational Logic Circuits: Digital logic, Boolean algebra, Boolean function, Canonical forms, Karnaugh Maps, Minimization of Boolean functions, Logic gates and their truth tables, Design methodologies, Combinational logic circuit design, Arithmetic and data handling logic circuits. Decoders, Encoders, Multiplexer, Demultiplexer.	CO 2	12	9
Flip Flop and Sequential Logic Circuits: Transistor Latch, NAND gate latch, NOR gate latch, D latch. Clock signals and Clocked FFs: Clocked SR, JK and D Flip-Flops, Master/Slave JK FF, timing diagram of different FFs, Edge-triggered and level-triggered timing diagrams. , Counters, registers, memory devices and their applications.	CO 3	9	18
Counters and registers: asynchronous counters and synchronous counters and their applications. Synchronous and asynchronous logic circuit design, State diagram; Mealy and Moor machines, State minimization and assignments, Pulse mode logic, Fundamental mode logic, Races and Hazards in sequential circuits. Memory unit. 555 Timer: Architecture of 555 Timer, different application of 555 timer, 555 as monostable and bistable Multivibrators.	CO 3	9	12

Recommended Books:

1. Ronald J. Tocci, Neal S. Widmer and Gregory L. Moss : Digital Systems.
2. M. Morris Mano : Digital Logic and Computer Design.
3. V. K. Jain : An Introduction to Switching Theory and Digital Electronics, Khanna Publishers, New Delhi

MATH 2201: Complex Analysis, Laplace and Fourier Transforms	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study complex analysis and its applications
- analysis laplace and fourier transforms and its applications

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn Complex plane, the extended plane and its spherical representation
CO 2	Learn Harmonic function. Complex Integration: Power series representation of analytic functions
CO 3	Evaluate basic properties and related theorems. Inverse Laplace transform and convolution
CO 4	Learn and apply Fourier series in various domains.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√											
CO3		√	√									
CO4		√	√	√	√							

Course Content:

Course Topics	CO	Contact Hours
Complex Analysis: Complex number system: Complex plane, the extended plane and its spherical representation (Riemann sphere). Complex function: Single and many valued function, Branch point, limit, continuity and differentiability of complex functions. Analytic functions: Necessary and sufficient conditions, Mobius transformation, Power series.	CO 1	9
Harmonic function. Complex Integration: Power series representation of analytic functions, zeros of analytic functions, Cauchy's theorem, Morera's theorem, Cauchy integral formula, Singularities and its classifications. Maximum modulus	CO 2	12

theorem, the homotopic version of cauchy's theorem and simple connectivity, the open Mapping theorem, Taylor's and Laurent series, Fundamental theorem of algebra, Rouches theorem. The argument principle, The Residue Theorem Contour integration. Conformal mapping, bilinear mapping. The application of the conformal mapping, Riemann Mapping theorem, Riemann zeta function, Analytic continuation, Riemann surface.		
Laplace Transform: Definition, existence and basic properties and related theorems. Inverse Laplace transform and convolution, Solution of linear differential equations with constant coefficients and linear systems.	CO 3	12
Fourier Series: Fourier integrals and applications. Fourier sine and cosine transforms, Complex Fourier transform, convolution theorem, Applications to boundary value problem.	CO 4	9

Recommended Books:

1. J. B. Conway : Functions of one complex variable
2. M. R. Spiegel : Theory and problems of Complex variables
3. Rajput: Mathematical physics
4. Spiegel: Laplace Transforms

STAT 2201: Theory of Statistics	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study different techniques of statistics
- analysis decision rule and test sampling

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn Fisher's Lemma, Study of 2-Distribution, T-Distribution and F-Distribution
CO 2	Learn Consistent estimates, unbiased estimates, Mean and variance of estimated
CO 3	Evaluate Statistical decisions, Statistical hypothesis: Critical region
CO 4	Learn and compute various tests of significance.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√											

CO3		√	√									
CO4	√	√	√									

Course Content:

Course Topics	CO	Contact Hours
Sampling Distribution: Fisher's Lemma, Study of 2-Distribution, T-Distribution and F-Distribution, Properties, uses and Applications, Distribution of sample correlation coefficient in the null case, Sampling Distribution of the Medians and Range.	CO 1	9
Elements of Point Estimations: Basic Concepts, Consistent estimates, unbiased estimates, Mean and variance of estimated Ideas of Efficiency, Principle of Maximum Likelihood, Illustration from Binomial, Poisson and Normal Distributions.	CO 2	9
Decision Rules: Statistical decisions, Statistical hypothesis: Critical region, best critical region, two types of errors, Procedure of test of hypothesis, most powerful test, and standard Errors.	CO 3	12
Test of Significance: Test of single mean and single variance, Comparison of two sample Means, proportions and variances, Bartlett's test for homogeneity of variances, Test for correlation and Regression coefficients, Exact test for 2x2 tables, Test for rxc tables, Three-way contingency tables, Large Sample Test of Significance, Non-Parametric Test, One Sample and two Sample Sign Test, Run Test and Rank Sum Test.	CO 4	12

Recommended Books:

1. R.N. Shill & S.C. Debnath : An introduction to the theory of Statistics
2. M.G. Mostafa :Methods of Statistics
3. Murry R. Spiegel : Theory and problems of Statistics
4. S.P. Gupta : Advanced Practical Statistics

CSE 2250: Viva-Voce
Credits: 0.75 Contact Hours: 0L+0P Hrs/Week

The viva-voce is designed to attain comprehensive knowledge about the whole semester along with that one can improve his/her attitude.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	The viva-voce is designed to attain comprehensive knowledge about the whole semester along with improvement of his/her communication and attitude.
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CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1										✓		✓

YEAR-3 TERM-1 B. Sc. ENGINEERING COURSES

CSE 3101: Computer Architecture and Organization	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- the basics of organizational and architectural design of a digital computer system.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn various hardware and software components of computers.
CO 2	Learn different design methods of processors, multiplier control unit, and CPU control unit.
CO 3	Evaluate different types of semiconductor memory, magnetic memory, and optical memory.
CO 4	Understand the system organization and design the memory and simple microcomputer system.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	✓											
CO2	✓											
CO3		✓	✓									
CO4		✓	✓		✓							

Course Content:

Course Topics	CO	Contact Hours
Concepts and Terminology: Digital computer components Hardware & Software and their dual nature, recent development, Role of Operating Systems (OS). Processor Design: Introduction: Processor organization, information representation, number formats; Fixed Point	CO 1	9

Arithmetic: Addition, subtraction, multiplication, division; ALU Design: Basic ALU organization, floating point arithmetic.		
Control Design: Hardwired control: Design methods, multiplier control unit, CPU control unit; Basic concept of Micro programmed Control, Control memory optimization.	CO 2	9
Memory Devices and its Organization: Different types of semiconductor memory, magnetic memory, optical memory, virtual memory, memory hierarchies; High-speed Memories: Interleaved memories, caches, associative memories.	CO 3	12
System Organization: Communications: Introduction, bus control; IO Systems: Programmed IO, DMA and interrupts, IO processors. Application HDL for microcomputer design: Description of Adder, ALU by using HDL, implementation of a simple microcomputer system using HDL.	CO 4	12

Recommended Books:

1. John P. Hayes: Computer Architecture and Organization
2. Carl Hamacher, Zvonko Vranesic and Safwat Zaky: Computer Organization
3. William Stallings: Computer Organization and Architecture: Designing for Performance

CSE 3103: Compiler Design	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 3104: Compiler Design Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to introduce the major concept areas of language translation and compiler design

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn and explains Lexical Analysis, Syntax Analysis, Intermediate
CO 2	Design and implement a software system for backend of the compiler
CO 3	Evaluate the knowledge of patterns, tokens & regular expressions for solving a problem in the field of data mining

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											

CO2		√	√		√							
CO3		√	√		√							

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction: Introduction to Compiler, Compiler and Translator, Compiler Structure, Applications of Compiler Technology. Lexical Analysis: Token, Patterns and Lexeme, Parser and Symbol Table, Difficulties in Lexical Analysis, Error Reporting, Regular Expression and Regular Definition, Transition Diagrams, LEX. Syntax Analysis: CFGs, Ambiguity, Associativity, Precedence, Top Down Parsing, Recursive Descent Parsing, Transformation on the Grammars, Predictive Parsing, Bottom up Parsing, LR Parsers, YACC.	CO 1	12	12
Intermediate Code Generation: Intermediate Representations, Three-Address Code –Quadruples, Triples; Translation of Expressions, Type Checking, Control Flow – Short-Circuit Code, Jumping Code; Procedure Calls.	CO 2	9	9
Run Time Environments: Storage Organization, Activation Tree, Activation Record, Heap Storage Management, Garbage Collection. Code Generation: Issues, Basic Blocks and Flow graphs, Optimization of Basic Blocks, Register Allocation, DAG Representation of Programs, Code Generation from DAGs, Code Generation Algorithm, Peephole Optimization, Instruction Selection.	CO 3	9	9
Optimization: Sources of Optimization – Redundancy, Dead Code Elimination, Code Motion, Induction Variables; Data-Flow Analysis, Instruction-Level Parallelism.	CO 2 CO 3	12	12

Recommended Books :

1. Alfred V. Aho, Monica S. Lam, Ravi Sethi, and Jeffrey D. Ullman : Compilers: Principles, Techniques, & Tools, *Pearson Education*.
2. Allen I. Holub: Compiler Design in C, *Prentice-Hall*.

3. Keith Cooper (Author), Linda Torczon: Engineering a Compiler, *Morgan Kaufman, Elsevier*.

4. Ronald Mak: Writing Compilers and Interpreters: A Software Engineering Approach, *Wiley Publishing*.

CSE 3105: Numerical Methods	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 3106: Numerical Methods Lab	
Credits: 0.75	Contact Hours: 0L+3/2P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to study numerical methods, based upon sound computational mathematics
- to understand the basic algorithms underpinning computer predictions in modern systems science

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn and analyze errors in mathematical operations on the computer.
CO 2	Recognize major numerical methods and their merits and pitfalls.
CO 3	Estimate the accuracy in approximated numerical solutions.
CO 4	Use different software tools based on different numerical methods for solving range of problems.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√	√										
CO2	√	√	√									
CO3		√	√									
CO4				√	√							

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Approximations and Errors: Accuracy and Precision, Error Definitions, Round-Off Errors, Truncation Errors. Roots of Equations: Graphical Methods, The Bisection Method, The False-Position Method, Simple One-Point Iteration, The Newton-Raphson Method, The Secant Method.	CO 1	9	6
Systems of linear algebraic equations: Gauss Elimination, Solving Small Numbers of Equations, Naive Gauss Elimination, Pitfalls of Elimination	CO 2	9	6

Methods, Matrix Inversion and Gauss –Seidel, The Matrix Inverse, Error Analysis and System Condition.			
Curve Fitting: Linear Regression, Polynomial Regression, Multiple Linear Regression, Newton's Divided-Difference Interpolating Polynomials, Lagrange Interpolating Polynomials, Coefficients of an Interpolating Polynomials, Curve Fitting with sinusoidal Functions.	CO 3	12	6
Numerical Differentiation and Integration : The Trapezoidal Rule, Simpson's Rules, Integration with Unequal Segments, Romberg Integration, Gauss Quadrature, High-Accuracy Differentiation Formulas, Richardson Extrapolation, Derivatives of Unequally Spaced Data. Numerical Solutions of Ordinary Differential Equations: Euler's Method, Modifications and Improvements of Euler's Methods, Runge-Kutta Methods, Adaptive Runge-Kutta Methods. Introductory concept of eigen value and eigen vector	CO 4	12	3

Recommended Books:

1. Steven C. Chapra, Raymond P. Canale : Numerical methods for engineers.
2. S.S.Sastry : Numerical Differentiation and Integration of Introductory Methods of Numerical Analysis.

CSE 3107: Database Management Systems	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 3108: Database Management Systems Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- Prepare themselves for tomorrow's database designer.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn and apply Files, Databases and Database Management Systems
CO 2	Learn and apply Normal forms, Normalized Relations, Database performance and De-normalization
CO 3	Evaluate Data Definition Language and Data Manipulation Language
CO 4	Learn Failure classification, Storage structure, Recovery and atomicity

CO No	Program Learning Outcomes (PLO)
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	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√	√		√								
CO2	√	√	√	√								
CO3		√	√									
CO4	√	√			√							

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Concepts of database systems: Files and Databases, Database Management Systems; Transaction management, Structure of a DBMS, Applications. Entity-Relationship concepts: Entity types, Entity set, Attribute and key, Relationships, Relation types, Entity relationship, ER modeling, ER diagrams, Database design using ER diagrams, Enhanced Entity-Relationship (EER) model.	CO 1	9	9
Normalization: Normal forms, Normalized Relations and Database performance; Denormalization. Relational model: Structure of relational databases, Relational algebra, Relational algebra operations, Modification of the database, Introduction to views, Pitfalls in relational database design.	CO 2	12	12
SQL: Data Definition Language, Data Manipulation Language, Basics of SQL, Query designing in SQL using aggregate functions and nested queries, Embedded SQL, Triggers, Procedures; Indexes; Declarative Constrains and Database Triggers. Concurrency control: Lock based protocols, Timestamp based protocols, Validation based protocols, Deadlock.	CO 3	12	12
Recovery system: Failure classification, Storage structure, Recovery and atomicity, Logbased recovery, Recovery with concurrent transactions, Advanced recovery techniques, RAID model. Advanced database management systems: No SQL Systems, distributed systems, objectoriented System, Temporal, Database Security, Data	CO 4	9	9

Warehousing and Data Mining, Database Administration and Tuning.			
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Recommended Books:

1. Abraham Silberschatz, Henry F. Korth, S. Sudarshan : Database System Concepts
2. Morgan Kaufmann, Elsevier : Database Design – Know it all
3. Eric Redmond, Jim R. Wilson : Seven Databases in Seven Weeks - A Guide to Modern Databases and the NoSQL Movement
4. Thomas M. Connolly, Carolyn E. Begg. : Database Systems - A Practical Approach to Design, Implementation and Management

ECE 3101: Data Communication	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
ECE 3102: Data Communication Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to study different techniques of communication
- analysis different modes of communication

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn and understand Data communication concept, Analog and digital data, spectrum and bandwidth
CO 2	Learn and understand different types of noise, encoding
CO 3	Learn Frequency division multiplexing, time division multiplexing

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√	√										
CO2	√	√										
CO3	√	√										

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction: Data communication concept, Analog and digital data, spectrum and bandwidth, Transmission requirements, Data rate and channel capacity, Asynchronous and synchronous data transmission techniques, Error detection and	CO 1	9	9

correcting codes, Data compression algorithms, Transmission Media: Characteristics and applications of Twisted pairs coaxial cables and optical fibers.			
Noise: Definition, Classification of noise, Noise due to several sources, Signal to noise ratio	CO 2	9	9
Data Encoding: Data and signal, RZL, NRZL, Manchester and differential Manchester encoding, ASK, FSK, PSK, QPSK, modems, sampling theorem, pulse Code modulation techniques, Delta modulation. Modulation Theory: Definition, Types of modulation: AM, FM, Comparison of FM and AM.	CO 3	12	12
Data transmission technique: Frequency division multiplexing, time division multiplexing, wavelength division multiplexing, Circuit switching, Packet switching, Hybrid switching. Optical fiber communication: Introduction, principle of light transmission in a fiber, modulation techniques.	CO 3	9	9

Recommended Books:

1. William Stallings : Data & Computer Communications, Prentice Hall Publication
2. Behrouz A. Forouzan : Data Communications and Networking, McGraw-Hill
3. John M. Senior : Optical Fiber Communications , Prentice-Hall of India Pvt Ltd

CSE 3150: Viva-Voce
Credits: 0.75 Contact Hours: 0L+0P Hrs/Week

The viva-voce is designed to attain comprehensive knowledge about the whole semester along with that one can improve his/her attitude.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	The viva-voce is designed to attain comprehensive knowledge about the whole semester along with improvement of his/her communication and attitude.
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CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1										✓		✓

YEAR-3 TERM-2 B. Sc. ENGINEERING COURSE

CSE 3200: Project	
Credits: 1	Contact Hours: 0L+2P Hrs/Week

Students will design and implements projects with proper documentation as assigned by teacher to attain industry skill.

CSE 3201: System Analysis and Design	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study information techniques and design
- analysis different types of system

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn information systems, general design consideration of information systems
CO 2	Learn Systems planning, design and development
CO 3	Evaluate system testing , quality assurance and deployment
CO 4	Learn and analysis Case studies of various information systems

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	✓											
CO2		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CO3				✓	✓	✓	✓	✓	✓	✓	✓	✓
CO4	✓	✓	✓						✓	✓	✓	

Course Content:

Course Topics	CO	Contact Hours
Introduction: Introduction to information systems, general design consideration of information systems.	CO 1	9

Overview: system concepts and the information systems environment, information needs, the concepts of MIS, the system development life cycle, the role of the systems analysis.		
Systems Analysis: Systems planning and the initial investigation, information gathering, the tools of structured analysis, feasibility study, cost benefit analysis. Systems Design: The process and stages of systems design, input/output and forms design, file organization and data base design.	CO 2	12
System Implementation: system testing and quality assurance, implementation and software maintenance, hardware/software selection, project scheduling and software, Security, disaster/recovery, and ethics in system development.	CO 3	12
Case study: Case studies of various information systems such as: Library management system, inventory system, voter identity management system, payroll system, etc.	CO 4	9

Recommended Books:

1. E.M. Awad: System Analysis and Design
2. P. Edwards: System Analysis & Design
3. J.G. Burch Jr., F.R. Strater and G. Grundnitski: Information Systems: Theory and Practice
4. G. Scott.: Principles of Management Information Systems

CSE 3203: Operating Systems	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 3204: Operating Systems Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to learn fundamental concepts of operating systems
- analysis of algorithms and technique of operating systems

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Understand how operating systems work and manage its hardware and software
CO 2	Able to program and/or modify system level functionalities of operating system including shell programming
CO 3	Evaluate Concepts of File and File System, Directory and File System Structure

CO 4	Learn Access Rights, Access Matrix, Program Threats, System Threats
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CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√		√								
CO3	√	√		√								
CO4	√	√						√				√

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction: Introduction to Operating System, Evaluation of Operating Systems and Computer Architectures, Operating System Structure, Virtualization, System Boot Process.	CO 1	9	9
Process management: Concept of Process and Thread, Operations on Processes, Inter-Process Communication, Process Scheduling, CPU Scheduling, Process and Thread Synchronization, Concepts of Deadlock. Memory Management: Background, Logical vs. Physical Address Space, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Demand Paging, Page Replacement, Thrashing.	CO 2	12	12
Storage Management: Concepts of File and File System, Directory and File System Structure, Directory and File System Implementation, Storage Allocation Methods to File, Free Space Management, Disk Management. I/O Management: I/O Hardware, Application I/O Interface, Kernel I/O Subsystem, Performance.	CO 3	9	9
Protection and Security: Basics of Protection and Security Problem, Access Rights, Access Matrix, Program Threats, System Threats. Case study: Case studies of Some Modern Operating Systems.	CO 4	12	12

Recommended Books:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne: Operating Systems Concepts, *Wiley Publisher*.
2. Andrew S. Tanenbaum : Modern Operating Systems, *Prentice-Hall*.
3. Thomas Anderson and Michael Dahlin: Operating Systems: Principles and Practice, *Recursive Books*.
4. Silberschatz, Abraham and Peter Baer Galvin and Greg Gagne: Operating system concepts with Java, *Wiley Publisher*.

CSE 3205: Web Engineering	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 3206: Web Engineering Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to introduce the major concept of developing web application

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Evaluation of the web, Categories of web application
CO 2	Web Application Components, Challenges of Web Application Development, Web Development Methodologies
CO 3	Design & implement a web application
CO 4	Learn E-Commerce Definition, Internet History and E-Commerce Development, Classification of E-Commerce

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√	√	√								
CO3		√	√		√	√	√	√	√	√	√	√
CO4	√											

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Principles and techniques: Introduction and perspective: Evaluation of the web, Categories of	CO 1	9	9

web application, Evaluation of Web engineering, Web development process, Knowledge and skills for web development, Web engineering resources portal (WEP):A reference model.			
Web Application Development Methodologies: Web Applications, Web Application Components, Challenges of Web Application Development, Web Development Methodologies, the Advantages and Disadvantages. Relationship Analysis: A Technique to Enhance Systems Analysis for Web Development: Generic Relationship Taxonomy, Conducting a Relationship Analysis, Relationship Analysis Model: Theoretical Basis, Relationship Analysis Process, Web Maintenance and Evolution, Techniques and Methodologies. Internet and World Wide Web application, HTML, SGML, XML, DTD, PHP, Java Script, CGI Programming etc.	CO 2	21	21
E-Commerce Basics: E-Commerce Definition, Internet History and E-Commerce Development, Classification of E-Commerce, E-Commerce Challenges, E-Commerce Opportunities, Payment Processing- Electronic Payment Issues, E-Cash, Credit Card Issues, Merchant Accounts, Online Payment Services, Transaction Processing.	CO 3 CO 4	12	12

Recommended Books:

1. *Web Engineering: Roger Pressman*
2. Achyut S Godbole & Atul Kahate : Web Technologies
3. David Kosiur: Understanding Electronic Commerce
4. Dave W. Mercer, Allan Kent, Steven D. Nowicki: Beginning PHP 5

CSE 3207: Digital Signal Processing	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 3208: Digital Signal Processing Lab	
Credits: 0.75	Contact Hours: 0L+3/2P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to learn digital signal processing including basic principles governing the analysis
- Design of discrete-time systems as signal processing devices.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn different signals and systems
CO 2	Apply the principles of discrete-time signal analysis to perform various signal operations
CO 3	Apply the principles of Fourier transform analysis to describe the frequency characteristics of discrete-time signals and systems
CO 4	Use computer programming tools to process and visualize signals

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√	√	√								
CO3		√	√	√								
CO4		√	√	√	√							

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction: Signals, systems and signal processing, classification of signals, the concept of frequency in continuous time and discrete time signals, analog to digital and digital to analog conversion. Discrete time signals and systems: Discrete time signals, discrete time systems, analysis of discrete time linear time invariant systems. Discrete time systems described by difference equations. Sampling and Reconstruction of signals: Sampling, A-D conversion, D-A conversion.	CO 1	9	6
The z-transform: Introduction, definition of the z-transform, z-transform and ROC of infinite duration sequence, properties of z-transform inversion of the z-transform, Frequency analysis of signals and systems: Frequency analysis of continuous time signals, Frequency analysis of discrete time signals, Properties of Fourier transform of discrete time signals.	CO 2	12	6
The Discrete Fourier Transform: The DFT, Properties of the DFT, Filtering method based on the DFT, Frequency analysis of signals using the	CO 3	12	6

DFT. Study of various types of filter. Fast Fourier Transform Algorithms: FFT algorithms, applications of FFT algorithm.			
Application of DSP: Speech processing, analysis and coding, Mat lab application to DSP.	CO 4	9	3

Recommended Books:

1. S Poornachandra, B Sasikala : Digital Signal Processing
2. S salivahana: Digital Signal Processing
3. Sanjit k. Mitra: Digital Signal Processing (A computer based approach)
4. J. G. Prokis: Digital Signal Processing

CSE 3209: Microprocessors and Assembly Languages	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 3210: Microprocessors and Assembly Languages Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to learn and understand assembly language
- to learn the major concept of major concept of microprocessor

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn Program structure and it components
CO 2	Evaluate Logic, Shift and Rotate Instruction, array and string
CO 3	Learn Microprocessors and Microcomputers, Evolution of microprocessors, Microprocessors applications
CO 4	Learn Control signals; I/O port organization
CO 5	Learn System design with different family microprocessors

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√										
CO3	√		√	√								
CO4	√											
CO5	√	√	√									

Course Content:

Course Topics	CO	Contact	Contact
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		Hours (Theory)	Hours (Lab)
Introduction of Assembly Language: Program structure and its components, few basic instructions, input/output instructions. Flag Register and Flow Control: The flag register, flow control instructions, conditional and unconditional jumps, branching and looping structures.	CO 1	9	9
Logic and Arithmetic Operation: Logic, Shift and Rotate Instructions, multiplication and division Instructions. Arrays and Data Structure: Arrays and related addressing modes, DUP operator, register indirect modes, Based and Indexed addressing modes, basic stack operations, procedures declaration, communication between procedures, calling a procedure. String Manipulation: The string instructions, director flag, moving a string, storing a string, Loading a string, scanning a string, comparing strings, substring operation..	CO 2	9	9
Microcomputer System: Microprocessors and Microcomputers, Evolution of microprocessors, Microprocessors applications. Intel 8086 Microprocessor: Internal architecture, register structure, programming model, addressing modes, instruction set, assembly language programming, condition tests, branches;	CO 3	12	12
Control structures; subroutine and parameter passage; macros I/O. Control signals; I/O port organization and accessing; 8086 family of chips, interrupts and interrupt Handling; Coprocessors.	CO 4	6	6
System design using 8086; an overview of Intel 80186, 80286, 80386, 80486 and Pentium microprocessors. RISC processors, 6800 and 68000 processors.	CO 5	6	6

Recommended Books :

1. Ytha Yu and CharlersMarut: Assembly Language Programming and Organization of the IBM PC, McGraw-Hill
2. Rafiquzzaman : Microprocessor and Microcomputer based System Design, Crc Press Publication
3. D. V. Hall : Microprocessors and Interfacing, McGraw-Hill

4. Y. Liu and G. A. Gibson: Microcomputer Systems: 8086/8088 Family, Prentice-Hall

CSE 3250: Viva-Voce	
Credits: 0.75	Contact Hours: 0L+0P Hrs/Week

The viva-voce is designed to attain comprehensive knowledge about the whole semester along with that one can improve his/her attitude.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	The viva-voce is designed to attain comprehensive knowledge about the whole semester along with improvement of his/her communication and attitude.
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CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1										✓		✓

YEAR-4 TERM-1 B. SC. ENGINEERING COURSES

CSE 4100: Project / Thesis	
Credits: 1.5	Contact Hours: 0L+3P Hrs/Week

Study and solution of problems in the field of Computer Science and Engineering.

N.B.: The project / thesis topic selected in this course is to be continued in the CSE 4200 course.

CSE 4101: Software Engineering	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 4102: Software Engineering Lab	
Credits: 0.75	Contact Hours: 0L+3/2P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to process and design techniques of software engineering

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn about software and its nature
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CO 2	Learn Software Design and its different techniques
CO 3	Learn the Management and its quality assurance, Software Cognitive fundamentals, Concepts of software reengineering and Web engineering
CO 4	Learn Software reliability metrics, software reliability specification, statistical testing and reliability growth modeling

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√	√									
CO3	√							√	√		√	
CO4	√							√				

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction: Introduction to software and its nature, software engineering methods, professional and ethical responsibility of a software engineer. Software Process Model: Different types of software process model and their implementations, costs of software engineering. Software Requirement Analysis: Software Requirements analysis and their applications, software Prototyping, Basic concepts of different formal software specification.	CO 1	12	3
Design of Software: Software Design and its different techniques, Software configuration managements. System structuring, control models, modular decomposition, domain-specific architecture. Software Testing: Software Validation and Verification: Verification and validation planning, Software's testing strategies and different type of testing techniques, Art of debugging.	CO 2	9	6
Software Quality Assurance: Management and its quality assurance, Software Cognitive fundamentals, Concepts of software reengineering and Web engineering.	CO 3	9	6
Advance Topics: Software reliability metrics, software reliability specification, statistical testing and reliability growth modeling, Use of CASE tools	CO 4	12	6

and technological support in engineering software, introduction to unified modeling language–UML.			
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Recommended Books:

1. Ian Sommerville : Software Engineering
2. Roger S. pressman : Software Engineering –A practitioner Approach

CSE 4103: Artificial Intelligence	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 4104: Artificial Intelligence Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to learn various agents in Artificial Intelligence.
- analysis of Artificial Intelligence algorithms, Methods for the design of different Artificial Intelligence algorithms

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn various agents
CO 2	Learn Approaches and issues in knowledge representation
CO 3	Evaluate Logics of non-monotonic reasoning
CO 4	Learn Planning with state space search - conditional planning-continuous planning
CO 5	Learn and execute in PROLOG programming

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√										
CO3		√	√	√	√							
CO4	√	√										
CO5	√	√	√	√	√							

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction: History of AI - Intelligent agents – Structure of agents and its functions - Problem spaces and search - Heuristic Search techniques – Best-first search - Problem reduction - Constraint	CO 1	6	6

satisfaction - Means Ends Analysis.			
Knowledge Representation: Approaches and issues in knowledge representation- Knowledge - Based Agent- Propositional Logic – Predicate logic – Unification – Resolution - Weak slot - filler structure – Strong slot - filler structure. .	CO 2	9	9
Reasoning under uncertainty: Logics of non-monotonic reasoning - Implementation- Basic probability notation - Bayes rule – Certainty factors and rule based systems-Bayesian networks – Dempster - Shafer Theory - Fuzzy Logic.	CO 3	9	9
Planning and Learning: Planning with state space search - conditional planning-continuous planning - Multi-Agent planning. Forms of learning - inductive learning - Reinforcement Learning - learning decision trees - Neural Net learning and Genetic learning	CO 4	6	6
AI programming languages: Introduction to PROLOG, knowledge representation, domain, predicate, clauses, database, back tracking, unification, list, and compound object using prolog. Introduction to selected topics in AI: Neural Networks, Expert system, Robotics and Fuzzy logic	CO 5	12	12

Recommended Books:

1. Elaine Rich, Kevin Knight and Shivashankar B.Nair : Artificial Intelligence, Tata McGraw-Hill
2. Stuart J. Russell and Peter Norvig : Artificial Intelligence: A modern Approach, Pearson Education Asia
3. Nils J. Nilsson : Principles of Artificial Intelligence, Narosa Publishing House
4. L. H. Tsoukalas and R. E. Uhrig : Fuzzy and Neural Approaches in Engineering

CSE 4105: Digital Image Processing	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 4106: Digital Image Processing Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to study the basic idea in Digital Image Processing
- analysis of different algorithms, Methods for Digital Image Processing

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn Image and Digital Image Processing
CO 2	Learn and apply Image Enhancement in the Spatial Domain & Frequency domain
CO 3	Understand and evaluate Image Restoration, color image processing and image compression.
CO 4	Learn and apply morphological image processing
CO 5	Learn and apply segmentation and recognition

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√	√	√								
CO3		√	√	√	√	√	√	√				
CO4	√	√	√	√	√	√	√	√	√			
CO5	√	√	√	√	√	√	√	√	√		√	

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction and Fundamental to Digital Image Processing: What is Digital Image Processing, Origin of Digital Image Processing, Examples that use Digital Image Processing, Fundamental steps in Digital Image Processing, Components of Digital Image Processing System, Image sensing and acquisition, Image sampling, quantization and representation, Basic relationship between pixels.	CO 1	6	6
Image Enhancement in the Spatial Domain & Frequency domain: Background, Basic gray level transformation, Histogram processing, Basics of spatial filtering, Smoothing and Sharpening Spatial filters, Introduction to Fourier Transform and the Frequency Domain, Discrete Fourier Transform. Smoothing and Sharpening Frequency-Domain filters.	CO 2	9	9
Image Restoration: Image Degradation/Restoration Process, Noise models, Restoration in presence of noise, Inverse Filtering,	CO 3	9	9

Minimum Mean Square Filtering, Geometric mean filter, Geometric transformations. Color Image Processing: Color Fundamentals, Color models, Basis of full color image processing, Color transformations. Image Compression: Fundamentals, Image compression models, Error free compression, Lossy compression.			
Morphological image processing: Preliminaries, Dilations and Erosion, opening and closing, Some basic morphological algorithms. Image Segmentation: Detection of Discontinuities, Edge linking and boundary detection, Thresholding, Region oriented segmentation.	CO 4	6	6
Representation, Description and Recognition: Representation-chain codes, polygonal approximation and skeletons , Boundary descriptors-simple descriptors, shape numbers, Regional descriptors- simple, topological descriptors, Pattern and Pattern classes-Recognition based on matching techniques.	CO 5	12	12

Recommended Books:

1. Rafael C. Gonzalez & Richard E. Woods : Digital Image Processing
2. A. K. Jain: Fundamentals of Digital Image Processing
3. Mark S. Nixon & Albert S. Aguado: Feature Extraction and Image Processing
4. William K. Pratt: Digital Image Processing

HUM 4101 Sociology and Bangladesh Studies	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to learn about socioeconomic condition in Bangladesh
- to learn different aspect like freedom fight, government and legislatures

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn Sociological perspective
CO 2	Learn Origin of the 'Bangladesh'
CO 3	Learn about Bangabandhu Sheikh Mujibur Rahman
CO 4	Learn Constitution of the people's republic of the Bangladesh

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1								√	√	√		√
CO2								√	√	√		√
CO3								√	√	√		√
CO4								√	√	√		√

Course Content:

Course Topics	CO	Contact Hours
Sociology: Sociological perspective: definition, nature, scope and importance of sociology; Sociology and scientific approach: methods of social research, stages of social research; Primary concepts of sociology: society, community, association, institution, group; Social evolution: stages in the evolution of human civilization; Culture: definition, characteristics, culture contents (material and non-material), cultural lag, culture and civilization; Industrial revolution: the growth of capitalism, features and social consequences, socialism; Some current social problems: crime, deviance, juvenile delinquency, youth unrest; Technology and society: effects of technological factors on social life.	CO 1	9
The Emergence of Bangladesh: Origin of the name 'Bangladesh', Language Movement (1948-1952), The United Front 1954: The Rise of the Bengali Nationalism, Quest for autonomy and Self-determination: Six Point Programme, Prelude to the War of Liberation in 1971: Anti-Ayub Movement and Mass upsurge in 1969, Election of 1970.	CO 2	9
The Liberation War of Bangladesh 1971: The great speech of Bangabandhu Sheikh Mujibur Rahman on 7 th March 1971, Military crackdown and genocide, Formation of Bangladesh (Mujibnagar) Government and the Proclamation of Independence Order, significance of the PIO, The Liberation War : March 26, 1971 – Dec 16, 1971, Regional and Global reactions: Role of India, the USA, the USSR and the UN.	CO 3	12
Government of Bangladesh: Constitution of the people's republic of the Bangladesh 1972, Power and functions of the president and prime minister, Judiciary system of Bangladesh: power and functions of the High court and Appeal division, Administrative system of Bangladesh: historical background and emphasis on local government system.	CO 4	12

Recommended Books:

1. F.R.Khan : Principles of Sociology
2. Anthony Giddens : Sociology
3. R. C. Majumdar (ed.), The History of Bengal, vol. I
4. Jadunath Sarkar (ed.), The History of Bengal, vol. II
5. Sirajul Islam (ed.), History of Bangladesh 1704-1971, vols. I & II
6. Government of Bangladesh, Constitution of Bangladesh

CSE 4150: Viva-Voce

Credits: 0.75

Contact Hours: 0L+0P Hrs/Week

The viva-voce is designed to attain comprehensive knowledge about the whole semester along with that one can improve his/her attitude.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	The viva-voce is designed to attain comprehensive knowledge about the whole semester along with improvement of his/her communication and attitude.
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CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1										✓		✓

OPTIONAL-I COURSES

CSE 4107: Computer Simulation and Modeling

Credits: 3.0

Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study Computer Simulation design techniques
- analysis of Modeling in Computer technology

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn and apply Simulation modeling
CO 2	Learn and design techniques for discrete-event simulation
CO 3	Evaluate the valid and credible simulation models

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√	√										
CO2	√	√	√	√	√			√				
CO3		√	√	√	√							

Course Content:

Course Topics	CO	Contact Hours
Simulation modeling basics: systems, models and simulation; Classification of simulation models; Steps in a simulation study.	CO 1	14
Concepts in discrete-event simulation: event-scheduling vs. process-interaction approaches, time-advance mechanism, organization of a discrete-event simulation model; Continuous simulation models; Combined discrete continuous models; Monte Carlo simulation; Simulation of queuing systems.	CO 2	14
Building valid and credible simulation models: validation principles and techniques, statistical procedures for comparing real-world observations and simulated outputs, input modeling; Generating random numbers and random variates; Output analysis. Simulation languages; Analysis and modeling of some practical systems.	CO 3	14

Recommended Books:

1. Raj Jain : The Art of Computer Systems Performance Analysis
2. Trivedi, K.S : Probability and Statistics with Reliability, Queueing and computer science Applications
3. Law A.M, and Kelton, W.D : Simulation Modeling and Analysis

CSE 4109: Multimedia Technology
Credits: 3.0 Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study about different Multimedia techniques and design

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn various Multimedia systems and Operating Issue
CO 2	Learn techniques for Networking issues in multimedia

CO 3	Learn and apply about Security issues in multimedia
CO 4	Learn and apply Networked virtual environment(NVE)

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√		√	√		√					
CO3	√	√	√	√	√	√	√					√
CO4	√	√										√

Course Content:

Course Topics	CO	Contact Hours
Multimedia systems: introduction; Coding and compression standards; Architecture issues in multimedia. Operating systems issues in multimedia: real-time OS issues, synchronization, interrupt handling. Database issues in multimedia: indexing and storing multimedia data, disk placement, disk scheduling, searching for a multimedia document.	CO 1	9
Networking issues in multimedia: Quality-of-service guarantees, resource reservation, traffic specification, haping and monitoring, admission control; Multicasting issues; Session directories; Protocols for controlling sessions;	CO 2	9
Security issues in multimedia: digital water-marking, partial encryption schemes for video streams. Multimedia applications: audio and video conferencing, video on demand, voice over IP.	CO 3	12
Networked virtual environment(NVE): Networked virtual environment overview; forms of distributed interaction; example systems; NVE technologies and challenges; origins of NVE.	CO 4	12

Recommended Books:

1. Ze-Nian Li and Mark S. Drew: Fundamentals of Multimedia
2. John Villamil-Casanova and Louis Molina: Multimedia: An Introduction
3. Tay Vaughan: Multimedia: Making It Work

CSE 4111: Basic Graph Theory
Credits: 3.0 Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study about basic of Graph Theory.
- analysis of algorithms, Methods for the design for efficient Graph algorithms

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn about Simple graphs
CO 2	Learn and apply different techniques for Complementary graphs
CO 3	Evaluate and apply different graph problem

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√		√								
CO3		√	√	√	√	√						

Course Content:

Course Topics	CO	Contact Hours
Simple graphs, digraphs, subgraphs, vertex-degrees, walks, paths and cycles; Trees, spanning trees in graphs, distance in graphs;	CO 1	14
Complementary graphs, cut-vertices, bridges and blocks, k-connected graphs; Euler tours, Hamiltonian cycles, Chinese Postman Problem,	CO 2	14
Traveling Salesman Problem; Chromatic number, chromatic polynomials, chromatic index, Vizing's theorem, planar graphs, perfect graphs.	CO 3	14

Recommended Books:

1. Russell Merris: Graph Theory
2. Reinhard Diestel: *Graph Theory*(Graduate Texts in Mathematics)

CSE 4113: Parallel and Distributed Processing	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study about Parallel processing and technique
- analysis of algorithms in Parallel and Distributed Processing technique.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn about Parallel Processing and architecture
CO 2	Learn and apply different techniques for Forms of Distributed Processing
CO 3	Evaluate the performances of Software and Network Strategy

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	✓											
CO2	✓	✓	✓	✓	✓							
CO3		✓	✓	✓	✓							

Course Content:

Course Topics	CO	Contact Hours
Parallel Processing: Importance, architecture, Hardware and software issues; Architectures for parallel processing - Classifications, Comparative Study of Different Architectures: hardware issues in parallel processing, parallel programming.	CO 1	14
Distributed Processing: Definition, Impact of distributed processing on organizations, pitfalls in distributed processing. Forms of Distributed Processing : Function distribution, Hierarchical distributed systems, Horizontal distributed systems, Strategy : strategies for distributed data processing, control of complexity, problem of incompatibility, centralization vs. decentralization, cost and benefit analysis;	CO 2	14
Design of Distributed Data: distributed data, location of data, multiple copies data, conflict analysis, database management, distributed databases and applications; Software and Network Strategy: Software strategy, the ISO seven layers, architectural interfaces, physical link control, network management etc.	CO 3	14

Recommended Books:

1. Timothy G. Mattson, Beverly A. Sanders, and Berna L. Massingill : Patterns for Parallel Programming.
2. Shaobai Zhang , Xiefeng Cheng : *Parallel distributed processing*

YEAR-4 TERM-2 B. Sc. ENGINEERING COURSES

CSE 4200: Project / Thesis	
Credits: 3.0	Contact Hours: 0L+6P Hrs/Week

Continuation of project / thesis topic undertaken in CSE 4100

CSE 4201: Computer Networks	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 4202: Computer Networks Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to the major concept of Computer Networks
- analysis Network Layer Protocols and its application

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn uses of computer networks and addressing
CO 2	Learn different Reference Model
CO 3	Design and implement different networks.

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√										
CO3		√	√	√	√				√		√	√

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Introduction: Definition, uses of computer networks, network topology, network media, network devices, different types of network: LAN, MAN, WAN etc.	CO 1	12	12

IP addressing: Classification of IP addressing, subnet mask, CIDR, private IP Address, public IP address, sub netting, VLSM etc.			
Network Model: OSI Reference Model, TCP/IT Reference Model, ATM Reference Model, functions of the layers of different models, Network Protocols working at different layers.	CO 2	12	12
Data Link Layer Design Issues and Framing: Character count, byte stuffing, bit stuffing, error detection: cyclic redundancy check, parity bit checking and correction: Hamming code, windowing protocols: go back N ARQ, selective repeat ARQ, elementary data link protocols, high-level data link control, point to point protocol, the medium access control sub-layer. Multiple Access: Random Access; ALOHA, CSMA, CSMA/CD, CSMA/CA, channelized access, CDMA, TDMA, FDMA, controlled access, reservation, polling, token passing, Ethernet, wireless LANs and Bluetooth. Switching: Circuit switching, packet switching, message switching, routing algorithms, virtual circuit and datagram, congestion control algorithms, quality of service, internetworking, internetworking devices etc. Network Layer Protocols: Address resolution protocol, internet protocol, internet control, message protocol, ipv6, routing information protocol, open shortest path first, border gateway protocol, user datagram protocol, transmission control protocol. Transport Layer: Services provided to the Session layer, Quality of service, Transmission Controls protocols, Connection management, Addressing, Establishing and Releasing Connection, Flow control and Buffering, Multiplexing, Transport layer in Public network and ARPANET. Session Layer: Services provided to the Presentation layer, Data exchange, Synchronization, Dialog and Activity management, OSI session service primitives. Presentation Layer: Data Compression techniques, Data Encryption Standard, Public key cryptography, Application Layer: File transfer Access and management, Electronic Mail, Client Server,	CO 3	18	18

Internet: Introduction to internet and intranets, internet protocols, Internet services and goals, Domain Name System (DNS) and Addresses, FTP, Telnet, World Wide Web (WWW), Internet Relay Chat (IRC), DHCP and BOOTP.			
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Recommended Books:

1. William Stallings : Data & Computer Communications, Prentice Hall Publication
2. Behrouz A. Forouzan : Data Communications and Networking, McGraw-Hill
3. Andrew S. Tanenbaum : Computer Networks, Prentice Hall of India Pvt. Ltd

CSE 4203: Computer Graphics	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 4204: Computer Graphics Lab	
Credits: 1.5	Contact Hours: 0L+3P Hrs/ Week

Course Objective: At the end of the course a student is expected to:

- to study the basics of Computer Graphics
- analysis of different algorithms, Methods of Computer Graphics

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn Graphic Package and technique
CO 2	Learn Graphic Display Processors
CO 3	Design and implements interactive graphics.
CO 4	Learn and implement through OpenGL and DirectX

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√	√										
CO2	√	√										
CO3		√	√	√	√							
CO4		√	√	√	√				√		√	

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
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Introduction: History, Application of Computer Graphics, Graphics I/O Devices and Types. Graphics Software Design: Survey of Desired Function, Toward a Universal Graphic Language, Display Files, Databases for Pictorial Applications. Graphics Techniques: Point-Plotting Techniques, Line Drawing, Geometric Transformations, Windowing and Clipping, Raster Graphics.	CO 1	9	9
Hardware for Computer Graphics: Typical Small and Large System, Graphic Terminals, Plotters, Graphic Display Processors, Device Independent Graphics Systems. Graphics Software: A Simple Graphic Package, Segmented Display Files, Geometric Models, Picture Structure.	CO 2	12	12
Interactive Graphics: Input Techniques, Event Handling, Three-Dimensional Graphics, Curves and Surfaces, 3-D Transformation. Hidden Surface Problem: Back Face Removal, Hidden-Line Removal Curved Surfaces, Describing Points, Lines And Polygons, Some Hints For Building Polygonal Models, Color Perception, RGBA and Color Index Mode, Dithering, Blending, 3-D Blending With The Depth Buffer, Antialiasing, Fog, Fog Equations, The OpenGL ARB.	CO 3	12	12
API Specifies: Data Types, Function Naming Conventions, Platform Independence, Drawing Shapes With OpenGL and DirectX. Drawing in Space: Points, Lines, Circle and Polygons. Co-ordinate Transformations: Understanding Transformations, Matrix Munching Projections, Matrix Manipulation Color Lighting and Materials, Texture Mapping.	CO 4	9	9

Recommended Books:

1. Donald Hearn, M. Pauline Baker: Computer Graphics with OpenGL
2. Zhigang Xiang, Roy A. Plastock, : Theory and Problems of Computer Graphics
3. James D. Foley , Andries van Dam , Steven K.. : Computer Graphics: Principles and Practice in C

CSE 4205: Interfacing and Microcontrollers	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week
CSE 4206: Interfacing and Microcontrollers Lab	

Course Objective: At the end of the course a student is expected to:

- to study about Interfacing in computing device
- to learn and analysis of different type of Microcontroller and its application.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn Hardware and software interfacing
CO 2	Learn and apply programmable timer, bus architecture in interfacing
CO 3	Understand and implement interrupt and DMA
CO 4	Learn Memory Terminology
CO 5	Learn and evaluate co-processor configurations and microprocessor

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√		√	√							
CO3	√	√		√	√							
CO4	√											
CO5	√	√	√	√	√							

Course Content:

Course Topics	CO	Contact Hours (Theory)	Contact Hours (Lab)
Microprocessor Based System Design: Hardware and software interfacing in microcomputer system design, hardware and I/O design, building, debugging, testing and linking program modules, programming EPROM.	CO 1	9	3
Interfacing Components: 8284A programmable timer, bus architecture, bus timing, 8286 transceiver device, 8282 latches, 8288 bus controller, characteristics of memory and I/O interface, synchronous and asynchronous communication, serial I/O interface, 8251A communication interface, 8255A Programmable peripheral Interface.	CO 2	9	6
Interrupt System: Sources of interrupt, types of interrupt, handling interrupt request, interrupt vector and table, 8259A priority interrupt controller, daisy chain.	CO 3	9	6

I/O Controller and Peripheral Components: Interfacing ICs of I/O Devices, I/O ports, Programmable peripheral interface, DMA controller i.e. 8237A DMA Controller, interrupt controller, communication interface, interval timer, etc.			
Memory Device: Memory Terminology, CPU-Memory Connections, ROM Architectures and Time Diagram, Different type of ROM, Flash Memory, RAM Architectures and Time Diagram, Different type of RAM and Read/Write Cycle, Programmable Logic Device Architectures.	CO 4	6	3
Multi-processor configurations: co-processor configurations, Numeric data processor, I/O Processors. Analog and Digital Interface: Sensors, Transducers, D/A interface, A/D interface, AD and DA converters related chips, High power devices. Microcontrollers: Microcontroller basics, Introduction to microcontroller architecture, Microcontroller operations, Microcontroller families, 8051 microcontroller, microcontroller Programming, Arduino.	CO 5	9	3

Recommended Books :

1. Ytha Yu and Charlers Marut : Assembly Language Programming and Organization of the IBM PC, *McGraw-Hill*
2. Barry B. Brey : The Intel Microprocessors 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, and Pentium Pro Processor, Architecture, Programming, And Interfacing, *Pentic-Hall*
3. D. V. Hall : Microprocessors and Interfacing, *McGraw-Hill*
4. Y. Liu and G. A. Gibson : Microcomputer Systems: 8086/8088 Family, *Prentice-Hall*

HUM 4201: Industrial Management and Accounting	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study about different aspects in Industrial Management
- to understand and learn about Accounting

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn Management and organization
CO 2	Learn and analysis Transportation and storage
CO 3	Learn Basic accounting principles
CO 4	Evaluate the performances of algorithms
CO 5	Preparation of cost sheet

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√	√									
CO3		√	√									√
CO4	√	√										
CO5	√	√						√				

Course Content:

Course Topics	CO	Contact Hours
Industrial management: Administration, Management and organization, Authority and responsibility, Scientific management, Organization structure, organization chart, Span of control, Selection and recruitment of employees; training and its types, promotion, wage system and incentive; job-evaluation and merit rating, Plant layout, layout of physical facilities,	CO 1	9
Transportation and storage, Material handling, Maintenance, Maintenance policy, Production control in intermittent and continuous manufacturing industry, functions of production control, Purchasing procedures : Inventory-need and methods of control, Factors affecting inventory building-up, Economic lot size and reorder point.	CO 2	12
Accounting: Basic accounting principles, Objectives of Accounting, Transaction, Double Entry systems, Accounts and it's classification, Journals Cash book, Ledger, Trial Balance, Financial statement.	CO 3	9
Cost Accounts & objectives; Costs; Classification, Preparation of cost sheet, Cost volume profit (CVP) analysis, Standard costing, Process costing.	CO 4	12

Recommended Books:

1. Kiseso : Accounting Principles
2. Basu & Dus : Practice in accounting (Volume I)

3. Garrison : Managerial Accounting

CSE 4250: Viva-Voce	
Credits: 0.75	Contact Hours: 0L+0P Hrs/Week

The viva-voce is designed to attain comprehensive knowledge about the whole semester along with that one can improve his/her attitude.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	The viva-voce is designed to attain comprehensive knowledge about the whole semester along with improvement of his/her communication and attitude.
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CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1										✓		✓

OPTIONAL-II COURSES

CSE 4207: Cryptography and Network Security	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study about the basic of Cryptography techniques and its applications
- to study about the Network Security and its applications

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn various Security Architecture
CO 2	Learn techniques of cryptography
CO 3	Evaluate Authentication requirements and its algorithms
CO 4	Learn and design Network security

CO No	Program Learning Outcomes (PLO)
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	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√		√								
CO3		√	√	√	√			√				
CO4	√	√	√	√	√			√				√

Course Content:

Course Topics	CO	Contact Hours
Introduction: OSI Security Architecture - Classical Encryption techniques - Cipher Principles - Data Encryption Standard - Block Cipher Design Principles and Modes of Operation - Evaluation criteria for AES - AES Cipher - Triple DES - Placement of Encryption Function - Traffic Confidentiality	CO 1	9
Public key cryptography: Key Management - Diffie-Hellman key Exchange - Elliptic Curve Architecture and Cryptography - Introduction to Number Theory - Confidentiality using Symmetric Encryption - Public Key Cryptography and RSA.	CO 2	9
Authentication and hash function: Authentication requirements - Authentication functions, Message Authentication Codes - Hash Functions - Security of Hash Functions and MACs - MD5 message Digest algorithm - Secure Hash Algorithm - RIPEMD - HMAC Digital Signatures - Authentication Protocols - Digital Signature Standard.	CO 3	12
Network security: Authentication Applications: Kerberos - X.509 Authentication Service - Electronic Mail Security - PGP - S/MIME - IP Security - Web Security. System level security: Intrusion detection - password management - Viruses and related Threats - Virus Counter measures - Firewall Design Principles - Trusted Systems.	CO 4	12

Recommended Books :

1. William Stallings : Cryptography and Network Security, *Prentice Hall Inc.*
2. Wadlow : Process of Network Security, The Designing and Managing a Safe Network, *Addison-Wesley*

CSE 4209: VLSI Design	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to understand about VLSI basic and its applications

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn about Microelectronics and MOS Technology
CO 2	Learn Overview of fabrication process
CO 3	Evaluate the performances of Bi-CMOS circuits
CO 4	Learn and evaluate Design and Test-Ability

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2	√	√		√								
CO3		√	√	√	√							
CO4	√	√	√	√	√							

Course Content:

Course Topics	CO	Contact Hours
VLSI design methodology: top-down design approach, technology trends. MOS technology: Introduction to Microelectronics and MOS Technology, Basic Electrical Properties and Circuit Design Processes of MOS and Bi CMOS Circuits,, MOS, NMOS, CMOS inverters, pass transistor and pass gates, DC and transient characteristics.	CO 1	12
Overview of fabrication process: NMOS, PMOS, CMOS, Bi-CMOS process. NMOS and CMOS layout: Color plate Stick diagram, and design rules. CMOS circuit characteristics: Resistance and capacitance, rise and fall time, power estimation.	CO 2	9
Introduction to Bi-CMOS circuits: Shifter, an ALU Sub-System, adder, counter, multipliers, multiplexer. Data Path and memory structures, Buffer circuit design, DCVS Logic.	CO 3	9
Design and Test-Ability: Circuit partitioning, Floor planning and placement, Routing, Practical Aspects of Design Tools and Test-Ability MOS Design, Behavioral Description, Structural Description, Physical Description and Design Verification.	CO 4	12

Recommended Books:

1. Douglas A. Pucknell, Kamran Eshraghian : Basic VLSI Design
2. Linda E. M. Brackendury : Design of VLSI Systems A Practical Introduction

CSE 4211: Wireless Communication	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study and understand the wireless mode of communications.

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn various Wireless Communication Systems
CO 2	Learn techniques for Cellular Mobile Systems
CO 3	Evaluate Multiple Access Techniques for Wireless Communication
CO 4	Learn and implement Wireless Networking
CO 5	Learn and design Intelligent Cell Concept And Application

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√	√	√								
CO3		√	√	√	√							
CO4	√	√	√	√	√							
CO5		√	√	√	√			√	√			

Course Content:

Course Topics	CO	Contact Hours
Introduction To Wireless Communication Systems: Evolution of mobile radio communications, examples of wireless comm. systems, paging systems, Cordless telephone systems, comparison of various wireless systems. Modern Wireless Communication Systems: Second generation cellular networks, third generation wireless networks, wireless in local loop, wireless local area networks, Blue tooth and Personal Area networks.	CO 1	9
Introduction to Cellular Mobile Systems: Spectrum Allocation, basic Cellular Systems, performance Criteria, Operation of cellular systems, analog cellular systems, digital Cellular Systems. Cellular System Design Fundamentals: Frequency Reuse, channel assignment strategies, handoff Strategies, Interference and system capacity, tracking and grade off	CO 2	9

service, improving coverage and capacity.		
Multiple Access Techniques for Wireless Communication: Introduction to Multiple Access, FDMA, TDMA, Spread Spectrum multiple Access, space division multiple access, packet ratio, capacity of a cellular systems.	CO 3	6
Wireless Networking: Difference between wireless and fixed telephone networks, development of wireless networks, fixed network transmission hierarchy, traffic routing in wireless networks, wireless data services, common channel signaling, ISDN (Integrated Services digital Networks), advanced intelligent networks. Wireless LAN Technology - IEEE 802.11 Wireless LAN Standard - Bluetooth.	CO 4	9
Intelligent Cell Concept And Application: Intelligent cell concept, applications of intelligent micro-cell Systems, in-Building Communication, CDMA cellular Radio Networks.	CO 5	9

Recommended Books :

1. William Stallings : Wireless Communications and Networks
2. Theodore S. Rappaport : Wireless Communications

CSE 4213: Computational Geometry	
Credits: 3.0	Contact Hours: 3L+0P Hrs/Week

Course Objective: At the end of the course a student is expected to:

- to study the technique of Computational Geometry and efficient algorithms of Computational Geometry

Course Outcome: Upon completion of the course, the students will be able to:

CO 1	Learn various models of Computation
CO 2	Analyze different techniques and algorithms for Computational Geometry
CO 3	Evaluate and apply of Computational Geometry

CO No	Program Learning Outcomes (PLO)											
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CO1	√											
CO2		√	√	√	√							

CO3		√	√	√	√				√		√	
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Course Content:

Course Topics	CO	Contact Hours
Introduction: Historical perspective, algorithmic background, geometric preliminaries, models of Computation, Geometric searching, point location problem and range searching problems,	CO 1	14
Divide and conquer, amortization, multi-dimensional search, space sweep, duality and randomization. Convex hulls. Proximity, Closest pair problem, Intersection and union of rectangles,	CO 2	14
Voronoi and Delaunay diagrams, arrangements of lines and points, Geometry of rectangles, hidden surface removal, polygon triangulation, art gallery theorems, shortest paths, and lower-bounds.	CO 3	14

Recommended Books:

1. M. d. Berg, O. Schwarzkopf, M. v. Kreveld and M. Overmars: Computational Geometry: Algorithms and Applications
2. F. P. Preparata and M. I. Shamos: Computational Geometry: An Introduction
3. J. O. Rourke: Computational Geometry in C

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